HOW TO MAKE BETTER DECISIONS WITH LIMITED BUDGETS

A dissertation

submitted by

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Abstract

Better decision making for problems with limited budgets

Dedicated to my son, Jacob, the most transformative part of my entire PhD journey.

Acknowledgements

Everybody

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Chapter 1

Executive Summary

Better decisions in public health with limited budgets

- Prediction error is the wrong goal
- It's difficult to train models on decision error
- The methods for training decision error are opaque and difficult to use.

First, in work published Epidemilogy we introduce BPR.

My second project, ICML we show how train models

In the third project, presented at OPTIMAL (I HOPE) we provide best practices for decision aware training.

Ultimately, using the tools developed in this thesis, better decisions for everyone.

Chapter 2

Introduction

How can we make better decisions about allocating limited budgets using machine learning?

Machine learning is increasingly used to answer decision-making questions in domains where resources are limited and uncertainty is high.

Throughout this thesis, we will consider the motivating example of a public health authority who wishes to invest resources in opioid-related overdose death prevention across its territory. In an ideal scenario all locations would receive adequate resources, but in practice, budgets are limited and interventions need to be prioritized for the areas of greatest need. This is a decision making problem: *given a budget to intervene in limited number K of locations, which locations should be prioritized for intervention?* Possible interventions include naloxone distribution (a life-saving drug which can reverse an overdose), outreach campaigns, or increased access to treatment options such as methadone. We make several simplifying assumptions throughout: the cost of an intervention is the same in every location (a reasonable assumption if the locations have equal population), and that an intervention is equally effective in every location (less realistic, but if reliable data indicating otherwise were available it could easily be incorporated).

Given this challenge, how should public health practitioners inform their decision making? A natural choice would be to study historical data, and allocate resources to the locations with the highest observed overdose death counts. Although we show that this is often an effective baseline strategy, increasingly machine learning models are being used to predict future overdose death counts, and these predictions are then used to inform resource allocation decisions. These models are trained as standard regression problems: given historical data on overdose death counts and other predictive features, can we predict future death counts? Typically, some years of data are heldout for evaluation, and the model is trained to minimize the error between predicted and obseredved death counts in these heldout years. Finally, the predictions from the trained model are used to solve the downstream optimization problem of maximizing the number of potential overdose deaths reached by intervention, subject to the budget constraint of only intervening in K locations. Somewhat trivially, this optimization problem is solved by simply selecting the K locations with the highest predicted death counts.

Even though this approach sounds intuitive, we see a subtle disconnect between how the machine learning model is trained and how those predictions are used. The model is trained to minimize average prediction error across all locations, but the decision problem requires correctly identifying which locations rank in the top K for predicted death counts. To expose this disconnect, you could consider adding 1000 to the predicted death count for all locations in the top K. This would cause the prediction error to increase dramatically, but the decision about where to allocate resources would remain unchanged. Beyond this cartoonish example, the fact that the model is trained to optimize for a different objective than the downstream decision can lead to worse decisions and worse outcomes. This problem can be espeically pronounced when studying fatal opioid-related overdoses, which are relatively rare events. Because of this, the prediction error can be dominated by the large number of locations with zero or low death counts, while the most important locations to get right are those with the highest death count.

The solution to this disconnect appears straightforward: instead of training the machine learning model to minimize prediction error, we should train it to maximize decision quality. This is the core idea behind a family of techniques which have variously been called Decision Aware [Chung et al., 2022], Decision-Focused Learning (DFL) [**?**], End-to-End Learning [**?**] [Tang and Khalil, 2024a], and Predict-then-Optimize (PtO) or sometimes Predict-and-Optimize (PnO) [Elmachtoub and Grigas, 2022], [Geng et al., 2024]. Unsurprisingly, optimizing for decision quality rather than prediction error can lead to better decisions [Elmachtoub et al., 2025b], especially when our models of the world are imperfect and misspecified.

Despite the simplicity of the idea of optimizing for decision quality, there are challenges and open research questions in the practical implementation of these techniques. The primary challenge arises from the discrete nature of the decision problem. Machine learning models are typically trained using gradient-based optimization methods, where model parameters are adjusted in the direction that most improves the training objective. In discrete decision problems, this is not possible. Small changes to model parameters almost always lead to the same decision; a change in predicted values that does not change which locations have the highest K values will not impact our decision at all. The decision function is piecewise constant with respect to the model parameters, and therefore has zero gradient almost everywhere.

To adress this differentiability challenge, a variety of methods have been proposed that proivde approimate objectives or approximate gradients that can be used to train decision-aware models. However, empircal evaluations of these methods are often incomplete, and comparisons between approaches are often inconsistent across different decision problems. No clear guidance exists for practitioners on which method to select for a given decision problem.

This thesis therefore asks the following central question: how can machine learning models be trained and evaluated so that they produce better decisions when allocating limited resources? In the following chapters, the thesis introduces three contributions that improve how decision-aware machine learning models are evaluated, trained, and benchmarked.

2.1 Thesis Overview

The rest of the thesis is organized as follows:

- In Chapter 3, I directly consider the problem of *evaluating* forecasts of opioid overdose deaths, and introduce a new metric called Best Possible Reach (BPR) that better captures the decision quality of these forecasts.

- In Chapter 4, I address the *training* of decision-aware machine learning models, and introduce a new method called Decision-Aware Maximum Likelihood (DAML) that jointly optimizes for decision quality and probabilistic maximum likelihood. I also demonstrate the method on problems beyond public health.
- In Chapter 5, I set out to *benchmark* different decision-aware learning methods. Here, demonstrate several shortcomings in existing benchmarks and provide *practical guidance* for both researchers studying these methods and practitioners looking to implement them.

Chapter 3

Spatiotemporal Forecasting of Opioid-related Fatal Overdoses: Towards Best Practices for Modeling and Evaluation

3.1 Introduction

The opioid overdose epidemic in the United States has resulted in over 450,000 deaths during the past eight years, with more than 80,000 fatal opioid-related overdoses during 2022, the highest yet in a single year.[FB Ahmad et al., 2023] Managing the opioid overdose epidemic requires a constellation of efforts ranging from substance use treatment programs offering medications for opioid use disorder (e.g., methadone, buprenorphine),[Leshner and Mancher, 2019, Mattick et al., 2014] harm reduction programs with provisions for overdose education and naloxone distribution, and comprehensive mental health and social support services.[Clark et al., 2014, Hawk et al., 2015, Peiper et al., 2019] Beyond the provision of harm reduction and healthcare services, it is critical for policymaking to address the ever-evolving substance use environment and plan for targeted interventions.

There has been considerable variation in the availability of different types of opioids and the consequent increase in opioid use disorder and opioid-related fatal overdoses in the past two decades. The current fatal opioid overdose epidemic has been characterized by four waves.[Ciccarone, 2019, 2021] In the early 2000s, prescription opioids drove overdoses. Then, heroin-related deaths surged post-2010, followed by a fentanyl spike in 2013.[Und, 2023] This culminated with the fourth wave of combined stimulant and fentanyl-related overdoses.[Ciccarone, 2021] These shifts in supply accompanied changes in social and ecological conditions, impacting substance use behaviors in varying ways across geographic regions.[Friedman et al., 2020, Stopka et al., 2023] Hence, it is critical to examine local spatiotemporal variation in opioid overdose outcomes, identifying the most-impacted areas and predicting future outcomes to inform preemptive public health responses.

A growing body of research[Marks et al., 2021c] has explored spatiotemporal variations in the opioid overdose landscape. Yet *forecasting* approaches are in a nascent stage and there are few prediction studies at the population level[Borquez and Martin, 2022]. Other research focuses on patient-specific risk prediction [Lo-Ciganic et al., 2021, 2022, Tseregounis and Henry, 2021], assuming access to detailed, person-level demographic and medical history data. However, there are immense challenges in compiling rich datasets for person-level analysis. Most state-level public health authorities may not have access to the data and technical resources to conduct individual predictive modeling. Analyses focused on population-level predictions that solely depend on more readily-available aggregated data have the potential to be more easily adopted by public health authorities with limited resources.

While a number of prior studies have identified historical overdose "hotspots"[Acharya et al., 2022, Herlands et al., 2018, Lu et al., 2023], fewer studies have forecasted future spatiotemporal overdose spikes. Research that focuses on hotspots often assumes that identified clusters represent the locations where the highest needs will exist in the future. In our analyses, we show that this assumption does not always hold. Intervention and policies that rely on such findings may be acting on lagged measures of opioid burden, thereby limiting the effectiveness of interventions. Existing research also spans a broad range of spatial and temporal resolutions. In geographic space, studies range from coarser county-level analysis[Acharya et al., 2022], to finer analyses based on ZIP Codes, census tracts, or census block groups[Allen et al., 2023]. Temporally, studies range in focus across yearly aggregated[Acharya et al., 2022] data, quarterly [Heuton et al.] , or weekly data[Ertugrul et al., 2019a].

The overarching goal of our study is to help public health departments make short-term forecasts of future overdose events to enable planning of geographically and temporally targeted interventions that are cognizant of limited resources and needed intervention efficiencies. We focus specifically on development of forecasts at a fine spatial scale (census tracts) at annual intervals, which we selected to match the decision-making needs of public health agencies. In order to understand the role of different forecasting models and evaluation metrics on different communities, our evaluations cover two distinct catchment areas. First, we study Cook County, Illinois, covering over 5 million residents of Chicago and its surroundings, where we forecast death events across 1328 populated census tracts from years 2015-2022 through analysis of publicly available data. Second, we study the state of Massachusetts, where we forecast fatal overdose events across 1620 census tracts representing over 6 million residents from 2001-2021. These locations were selected based on data availability, and to demonstrate the impact of model and metric choice at multiple locations.

To establish best practices for modeling and evaluation, we carefully compare different modeling approaches and performance metrics in each catchment area. We implement a comprehensive set of existing models – including heuristic baselines, statistical models,[Allen et al., 2023, Bauer et al., 2023a, Ertugrul et al., 2019a, Marks et al., 2021c] and neural networks[Ertugrul et al., 2019a]. We then assess the opioid-related fatal overdose forecasts they produce for both Cook County and Massachusetts at the census-tract-level at annual timeframes. We compute widely-used error-based performance metrics and introduce a new intervention-focused performance metric. Our Python-based software is available for other researchers to reuse and build upon: https://github.com/tufts-ml/opioid-overdose-models.

3.2 Methods

3.2.1 Data Sources and Preparation

To assess models, we assembled two datasets suitable for forecasting opioid-related fatal overdoses annually at the census tract level. Our relatively coarse annual temporal scale was chosen to match the frequency at which decision-makers might set new priorities and at which new reliable data become available. We chose the census tract spatial scale due to its potential for targeted interventions at a sub-municipality level. Each census tract by design contains a mean count of 4000 people (with a range of 1200-8000 [US Census Bureau]) . For many (but not all) interventions, costs scale with population size, and thus the cost of deploying an intervention in any tract is roughly uniform.

3.2.1.1 Data source 1: Cook County, Illinois

We obtained fully de-identified data from the Cook County Government Medical Examiner Case Archive[Med] for opioid-involved overdose deaths from August 2014 (the first date records are available) to May 2023. These data contained every fatal incident under the medical examiner's jurisdiction that was determined to have any opioid as a primary cause. We used the provided incident latitude and longitude to map each overdose fatality to one of 1328 census tracts. Because the underlying data is in the public domain, we make our processed Cook County data available in our shared repository.

3.2.1.2 Data source 2: Massachusetts

We obtained death certificate data from the Massachusetts Registry of Vital Records and Statistics for opioid-involved overdose deaths between 2001 and 2019. These deaths were defined as unintentional, intentional, and undetermined drug poisonings containing an opioid code (ICD-10 codes T.40.0-T40.4, or T40.6) as a "multiple cause of death". Each fatal overdose is linked to a calendar date as well as a residential street address. Decedent addresses for the place of residence at the time of the fatal overdose were geocoded, assigning latitude and longitude measures to each event that was then mapped to one of 1620 census tracts using the 2020 census tract boundaries.

3.2.1.3 Dataset Preparation

For each dataset, we computed the observed number of fatal overdose events $y_{s,t}$ at time unit t for individuals residing in spatial tract s. We employed open tools [Freeman] that utilize the US Census Geocoding API to map locations (street address or latitude/longitude) to its corresponding census tract, using the tract boundaries for the states of Massachusetts and Illinois defined by the U.S. Census Bureau in 2020.

In each dataset, a uniform set of covariates is available as input for prediction models. At each time period t and spatial region s, we provide the history of fatal overdose counts from previous times in that region, as well as the spatial location (numerical latitude and longitude of the tract), and timestamp (numerical time, measured in years since the first available year for that dataset). Further, for each census tract s at time t, an optional covariate vector represents social vulnerability across socioeconomic status, age-related demographics, minority status, housing, and composite dimensions using percentile ranking across all census tracts in that state. These features stem from the five dimensions of the Social Vulnerability Index (SVI)[CDC, 2022] published between 2000 and 2018 for every U.S. state. Published tract values are updated every five years; we selected the closest value to each time period t. These SVI features were chosen for their simplicity and portability, mirroring the role of higher-dimensional socioeconomic covariates in previous studies [Bauer et al., 2023a, Marks et al., 2021a].

3.2.2 Metrics

To evaluate model forecasting accuracy against observed mortality in a test period, suitable performance metrics were essential. Our study considered both commonly-used error-based metrics and a new intervention-focused metric.

3.2.2.1 Error-based metrics

Model performance is often assessed via summary statistics of the errors between predicted and observed mortality across all S spatial regions in the test period.

Within this category, two common metrics are Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), both defined in Equation 1 below. RMSE calculates the square root of the average squared errors, while MAE computes the average of absolute errors.

$$\text{RMSE} = \sqrt{\frac{1}{S}\sum_{s=1}^{S}(y_s - \widehat{y_s})^2}, \quad \text{MAE} = \frac{1}{S}\sum_{s=1}^{S}|y_s - \widehat{y_s}| \tag{3.1}$$

Both RMSE and MAE have been concretely used as the primary metrics to assess opioid overdose forecasting[Bauer et al., 2023a, Ertugrul et al., 2019a, Sumetsky] . RMSE is more sensitive to outliers because it squares large errors.

3.2.2.2 Intervention-focused metric

In our intended use case of mitigating the opioid crisis, stakeholders at a public health agency could use a forecasting model to select a targeted subset of all possible census tracts in which to deploy an intervention in the near future. We assume these actors have a limited budget, allowing intervention deployment in a maximum of K of the S regions in their jurisdiction. For a given model, we can obtain its recommended set of K regions, which we refer to as the intervention set I, in two steps. Step 1: predict mortality counts for all S regions in the test period. Step 2: identify the K regions with the K highest predictions (breaking ties at random), and store these as the recommended set I.

To evaluate such intervention recommendations, we need new metrics. The error-based metrics defined earlier (RMSE or MAE) aggregate error for all S regions equally. They do not directly measure if a forecast's guess of the K highest-risk areas specifically aligns with the actual areas with highest mortality. We thus wish to design a metric better aligned with how stakeholders will determine and assess intervention priorities. We suggest a model is favorable if, during the test period, the total count of fatal overdose events in its recommended set I of K tracts is as large as possible. This would indicate the model is good at identifying where adverse events will occur, and thus increase the possibility that stakeholders could "reach" and hopefully mitigate these events via interventions targeted at the model's recommended tracts.

Our proposed metric, "best possible reach" (BPR), assesses a model's recommendations via a ratio of two numbers. First, the numerator counts how many opioid-related fatalities actually occurred in the model's recommended set I of size K. Second, the denominator counts the total number of opioid-related fatalities in the K regions that would be chosen with perfect hindsight of the actual count vector $y = [y_1, \ldots, y_S]$ of fatalities in the test period. Mathematically, we define BPR as follows

$$\text{BPR} = \frac{\sum_{k \in I} y_k}{\sum_{k \in \text{TopKInds(y)}} y_k} = \frac{\text{\# events in K regions picked by model}}{\text{\# events in actual K highest count regions}} \tag{3.2}$$

where TopKInds(y) denotes a function that returns the distinct indices of the K largest elements of vector y.

For public health applications, BPR holds a practical interpretation as the proportion of fatal overdose events the current model's interventions would reach, compared to the perfect foresight of future events. BPR's numerical value by definition will have a minimum of 0.0 and a maximum of 1.0. We typically convert the fractional BPR to a percentage ranging from 0-100%, denoted as %BPR. A higher %BPR value signifies a more effective model at deciding where to intervene. A value of 100% indicates perfect decision-making given a limited budget: there is no other set of K regions any model could have recommended that would reach more events.

Although independently developed by our team (see our preliminary workshop paper Heuton et al.), our proposed BPR metric closely resembles the metric suggested by a recent pre-registered trial[Marshall et al., 2022] and a later feasibility study[Allen et al., 2023] to evaluate opioid overdose forecasts in Rhode Island. The primary distinction lies in the denominator: our BPR sums only the top K indices, while the alternative includes all S regions. We prefer our definition due to %BPR's consistent range of 0-100%, with 100% representing a model that could not have made a better decision given the limited budget of K regions. In contrast, the alternative definition's maximum value fluctuates based on observed data in the test period, making it difficult to know if another model could have done better.

3.2.3 Models

Below we define a range of possible forecasting methods that can be used for our where-to-intervene prediction task. All methods are trained and evaluated via a common protocol using the same provided splits (train/validation/test) of the two available datasets (Cook County and Massachusetts). Thorough details about model fitting and hyperparameter tuning are provided in the Supplement. This study was reviewed by the Tufts University Health Sciences Institutional Review Board and deemed to be non-human subjects research.

3.2.3.1 Simple Baseline Models

We study several easy-to-implement baseline models to highlight their comparative strengths. Public health practitioners seeking data-driven allocation of scarce intervention resources without sophisticated modeling could easily use these approaches.

Our first baseline, dubbed *all zeroes,* predicts uniformly across all S tracts that zero fatal overdoses will occur in the test period. This model, by definition, ranks all tracts as equally high-risk, so for metrics like BPR that require a set of K recommended regions we report an average over many samples of K distinct regions selected uniformly at random.

Our second baseline, known as *last year,* predicts that the mortality at the next year in a specific location will mirror the mortality observed at the most recent recorded year for that location.

The final baseline we consider is a *historical average,* which predicts the next timestep's mortality as an average of all mortality counts observed over the preceding W timesteps. Here, the appropriate "look back" period length W serves as a model hyperparameter.

3.2.3.2 Complex Models

Next, we consider several more flexible models with parameters that can be fit to the data. The first is the *weighted historical* average: a weighted average of the previous W years of overdose events. This is more flexible than *historical average*, because each year's count is multiplied by a customized weight coefficient.

The next model we consider is a Generalized Linear Model (GLM) with a Poisson likelihood. This model assumes that fatal overdose count y for spatial tract s at time t is modeled by a Poisson distribution where the log of the mean parameter is a linear function of the covariate vector x for that tract and time:

$$y|\ x, \boldsymbol{\theta}\ \sim\ \text{Poisson}(\lambda\ = e^{\boldsymbol{\theta}^T x}) \tag{3.3}$$

GLMs have limited flexibility due to the assumption of a monotonic relationship between covariates and fatal event count. We thus also include a Gradient Boosted Trees[Friedman, 2001] model, a popular more flexible ensemble of regression trees. Previous studies of opioid forecasting [Allen et al., 2023, Schell et al., 2022] have used similar tree ensembles.

In addition, we include three spatially-sophisticated statistical models used in recent opioid overdose forecasting applications. First, we include a *Gaussian Process* model[Allen et al., 2023] for its ability to flexibly capture spatial and temporal correlations. We use similar covariance functions ("kernels") to prior overdose forecasting work (details in the supplement). Next, *Bayesian Spatio-Temporal (BST)* models[Bauer et al., 2023a] use a Markov Random Field to model inherent spatial and temporal trends. Thirdly, *NBSpLag* denotes a negative binomial regression model with spatially lagged features[Marks et al., 2021a], where each tract is informed by its spatial neighbors. In a variable selection experiment[Marks et al., 2021a], these spatially lagged covariates were found to be the most predictive features. Unlike previous evaluations of both *BST*[Bauer et al., 2023a] and *NBSpLag* models, our study compares to the rich set of baselines described above.

Finally, we include *CASTNet*[Ertugrul et al., 2019a], a neural network approach custom developed for opioid-overdose forecasting. Unlike previous methods, CAST-Net employs multi-head attentional networks that allow predictions at a given location to be informed by learned "communities" of regions.

3.2.4 Experimental Protocol

We applied each of the models described above separately to the Cook County, IL dataset and the MA dataset. In each case, we sought to use available historical counts of opioid-related fatal overdoses (together with other covariates described above) to predict future fatal overdose counts in each census tract. We further assessed how these predictions can be used to recommend where to intervene in the near future.

For training on a dataset, for all S regions we assemble covariate vector, fatality count pairs $(x_{s,t},\ y_{s,t})$ for each year in the training set (t = 2010-2018 for Massachusetts, 2015-2019 for Cook County). The historical covariates inside each $x_{s,t}$ vector can summarize the recent history of W previous years (W=10 for Massachusetts, W=5 for Cook County). Hyperparameters are chosen to maximize performance as assessed by BPR on a validation set of data from the year prior to evaluation (2019 for Massachusetts, 2020 for Cook County. Finally, models are evaluated on predictions for the final two years (2020-2021 in Massachusetts, 2021-2022 in Cook County).

From each model, we obtained predictions for each of the S tracts in each test year. We then computed each evaluation metric (RMSE, MAE, BPR) as well as an interval that quantifies our uncertainty in its precise value. Inspired by resampling methods for uncertainty quantification[Spa], for each test year we obtained 50 different without-replacement samples of 1370 of the 1620 locations in MA (1078 of the 1328 in Cook County, IL), and recorded all metrics of interest for each sample. Our reported intervals quantify the min-max range of these 50 samples. We chose the number of tracts retained in each catchment area so that each sample on average retains 85% of all fatal overdose events.

3.3 Results

Results from the experiments conducted on Massachusetts and Cook County data are summarized in Table 3.1 and Table 3.2, respectively. The best model(s) can vary depending on the chosen evaluation metric.

In both catchment areas, we see reasons to prefer our proposed BPR metric to alternatives when the goal is effectively prioritizing where to intervene. First, in Massachusetts, both the Gaussian Process (GP) and Bayesian Spatiotemporal model (BST) have top performance as assessed by MAE and RMSE. However, the BST has higher %BPR than the GP (62.0% compared to 58.2%). Interventions guided by the BST model would have the potential to preemptively identify 18 additional fatal overdoses per year in the top 100 census tracts. Similarly, in Cook County the Gradient Boosted Trees model with SVI covariates has superior MAE and RMSE to the GLM model, yet has worse %BPR (77.1% versus 79.4% for GLM). Interventions guided by the GLM model (preferred via the %BPR metric) could reach 15 more fatal overdose events annually in Cook County.

We also observe that while complex models like BST do well in both catchment areas, so does the simple *historical average* baseline and its weighted extension. In Massachusetts, *historical average* delivers a BPR and MAE that fall within the uncertainty intervals of the best performing models. The weighted extension's ultimate %BPR is so close to the top method (NPSpLag) that the difference amounts to identifying fewer than 2 additional overdose events annually. In Cook County, the *historical average* baseline delivers competitive scores (within the intervals of the best models) as assessed by all three metrics (BPR, MAE, and RMSE); the best model (BST) here would reach less than 1 additional overdose event annually.

3.4 Discussion

Our study's first contribution to the science of spatiotemporal forecasting of opioid-related overdose deaths is highlighting the need for extensive comparisons to a robust suite of simple baselines. This lesson matches reports[Hand, 2006, Rudin, 2019] from

Table 3.1: Comparison of fatal opioid-related overdose prediction models trained on Massachusetts decedent data from 2010-2019, then evaluated on data from 2020 and 2021.

Here BPR (higher is better) is our new intervention-aware metric, computed assuming an intervention budget for K=100 of 1620 possible tracts in Massachusetts. MAE (lower is better) and RMSE (lower is better) are common error-based metrics. For each metric, the best model mean is bolded, along with any other model whose uncertainty interval overlaps this mean (intervals are computed via resampling methods). The column titled "Total Overdoses Identified in top 100 tracts" contains the *true* number of observed fatal overdoses in the top 100 locations identified by the corresponding model.

Abbreviations: **BPR**: Best Possible Reach. **MAE**: Mean Absolute Error. **RMSE**: Root Mean Squared Error. **SVI**: Social Vulnerability Index covariates. **GLM**: Generalized Linear Model. **GP**: Gaussian Process. **NBSpLag**: Negative binomial regression with spatially-lagged features. **BST**: Bayesian spatiotemporal model

		Metric		
Model	Total overdoses identified in top 100 tracts	%BPR (K=100)	MAE	RMSE
All Zeros	124.1	25.1, (24.8-25.8)	1.24, (1.18-1.30)	1.92, (1.83-2.01)
Last year	254.5	51.8, (49.5-54.4)	1.09, (1.04-1.14)	1.58, (1.50-1.65)
Historical Average (4 year)	295.4	**59.8, (56.8-62.9)**	**0.92, (0.90-0.94)**	1.28, (1.24-1.33)
Weighted Historical Average	303.5	**61.4, (56.1-66.9)**	**0.94, (0.91-0.98)**	1.32, (1.27-1.38)
Poisson GLM	304.7	**61.6, (56.4-66.8)**	**0.95, (0.91-0.98)**	1.32, (1.25-1.39)
Poisson GLM +SVI	301.3	**61.0, (57.4-65.1)**	1.10, (1.04-1.16)	1.38, (1.30-1.47)
Gradient Boosted Trees +SVI	293.6	**59.5, (55.9-63.1)**	**0.92, (0.89-0.96)**	**1.24, (1.18-1.30)**
GP	287.2	58.2, (55.2-61.2)	**0.93, (0.91-0.95)**	**1.28, (1.23-1.34)**
CASTNet +SVI	268.0	54.4, (52.0-56.6)	1.07, (0.99-1.16)	1.47, (1.36-1.58)
BST: Bayesian spatiotemporal	301.0	**61.1, (58.6-63.3)**	0.95, (0.93–0.96)	1.26, (1.24-1.28)
BST + SVI	305.4	**62.0, (60.0-63.7)**	**0.92, (0.91-0.94)**	**1.23, (1.21-1.25)**
NBSpLag + SVI	305.4	**62.0, (60.3-64.0)**	**0.92 (0.89-0.95)**	1.31, (1.27-1.35)

Table 3.2: Comparison of fatal opioid-related overdose prediction models trained on Cook County, Illinois decedent data, from 2015 to 2020, then evaluated on data from 2021 and 2022.

Here %BPR (higher is better) is our new intervention-aware metric, computed assuming an intervention budget for K=100 of 1328 possible tracts in Cook County. MAE (lower is better) and RMSE (lower is better) are common error-based metrics. For each metric, the best model mean is bolded, along with any other model whose uncertainty interval overlaps this mean (intervals are computed via resampling methods). The column titled "Total Overdoses Identified in top 100 tracts" contains the *true* number of observed fatal overdoses in the top 100 locations identified by the corresponding model.

Abbreviations: **BPR**: Best Possible Reach. **MAE**: Mean Absolute Error. **RMSE**: Root Mean Squared Error. **SV**: Social Vulnerability covariates. **GLM**: Generalized Linear Model. **GP**: Gaussian Process. **NBSpLag**: Negative binomial regression with spatially-lagged features. **BST**: Bayesian spatiotemporal model

		Metric		
Model	Total overdoses identified in top 100 tracts	%BPR (K=100)	MAE	RMSE
All Zeros	137.0	21.7, (21.1–22.4)	1.37, (1.30–1.44)	2.46, (2.34–2.56)
Last year	466.0	73.9, (70.7–77.0)	1.07, (1.03–1.10)	1.66, (1.59–1.71)
Historical Average (4 year)	505.3	**80.1, (76.2–84.3)**	**0.94, (0.90–0.97)**	**1.44, (1.38–1.50)**
Weighted Historical Average	495.4	**78.6, (75.7–81.8)**	1.15, (1.11–1.19)	1.87, (1.76–1.96)
Poisson GLM	496.8	**78.8, (75.9–82.3)**	1.22, (1.08–1.29)	4.64, (1.65–5.92)
Poisson GLM + SVI	500.7	**79.4, (76.0–82.8)**	1.16, (1.11–1.20)	1.86, (1.77–1.94)
Gradient Boosted Trees +SVI	485.7	**77.1, (72.2–81.9)**	0.99, (0.95–1.05)	1.55, (1.42–1.68)
GP	477.2	**75.7, (69.9–80.5)**	1.03, (0.97–1.09)	1.63, (1.50–1.74)
CASTNet +SVI	472.6	75.2, (73.2–76.8)	1.01, (0.98-1.04)	1.53, (1.39-1.67)
BST: Bayesian spatiotemporal	505.9	**80.5, (78.9–82.1)**	1.00, (0.98–1.01)	**1.40, (1.36–1.43)**
BST + SVI	504.1	**80.2, (78.7–82.0)**	0.97, (0.95–0.98)	1.47, (1.43–1.51)
NBSpLag + SVI	502.2	**79.9, (78.1–81.6)**	**0.96, (0.93-0.98)**	**1.42, (1.37-1.48)**

across the sciences, especially efforts in health[Christodoulou et al., 2019, Nusinovici et al., 2020] and the social sciences[Salganik et al., 2020], that suggest advanced modeling techniques may not substantially outperform simpler baselines on some difficult prediction tasks. Our findings are similar in both the large state of MA and the far denser Cook County. In each catchment area, across both intervention-aware and error-based metrics, we found that a historical average baseline performed competitively (within the uncertainty bounds of top-ranked statistical models). The key to success here is careful selection of the number of recent years in the look-back period, following standard best practices for hyperparameter tuning.[Probst et al., 2018, Raschka, 2020] If this simple baseline model yields such high performance, it raises questions about the rationale for adopting more complex counterparts that require specialized expertise. Many prior overdose forecasting studies[Allen et al., 2023, Bauer et al., 2023a] completely omit such baselines, or often include only the poor performing ones such as the last-year[Marks et al., 2021a] model or a too-long historical average[Ertugrul et al., 2019a]. For all future studies of opioid overdose forecasting, we recommend including historical averages with tuned look-back periods.

Our second contribution, developed in parallel to contemporary work[Marshall et al., 2022], is a new metric – percentage of best possible reach (%BPR) – which evaluates predictions based on their utility for informing decisions about where to intervene. In both Massachusetts and Cook County, we demonstrate that using %BPR as an evaluation metric can lead to different model rankings and different recommendations of where to intervene than error-based metrics like RMSE, improving the total number of annual fatal overdose events that could be preemptively identified by 15 in IL and 18 in MA. This is an important finding, as we believe that intervention-aware metrics like %BPR more closely reflect how public health agencies wish to use forecasting models to inform their intervention strategies.[Allen et al., 2023]

Lastly, we emphasize that our study is designed to be reproducible and open to extensions by other researchers. We released the software for fitting all models and computing all metrics under a permissive open-source license (link in Introduction). We also released our cleaned version of the public-domain Cook County dataset as well as all preprocessing code. Historically, overdose forecasting studies have not often shared code and have focused on private bespoke datasets, often reasonably due to privacy issues around decedent data. Enabling diverse researchers to pursue a common prediction task, especially via the availability of a public dataset for evaluation, has been a key driver of progress in predictive modeling[Donoho, 2017].

3.4.1 Limitations

This study has several limitations. First, our findings come from only two places (Massachusetts and Cook County), and may not be generalizable to other counties, states, or public health jurisdictions. Cook County is predominantly urban, while Massachusetts is a large state with substantial urban, rural and suburban areas. The spatiotemporal trends in opioid-related mortality could thus be dramatically different in these two locations, necessitating different model rankings and intervention strategies. Furthermore, we acknowledge that not all Cook County deaths are reported to the Medical Examiner. The Medical Examiner's jurisdiction only covers specific fatalities for cause-of-death determination.

Second, there are limitations to our analysis of the proposed BPR metric. For simplicity, all results here assumed an intervention budget of K=100 census tracts. Different K values may lead to different method rankings. Our suggested BPR metric is intended for identifying where to intervene to relieve high overall burden. However, it does not directly prioritize the rate of change. Interventions aimed to reduce risk in communities that are at very high risk but do not already have a high burden may not be correctly identified using BPR.

Finally, other choices of covariates are possible. Our focus on a limited set of covariates, derived from the SVI of the American Community Survey, was an intentional choice to ensure the nationwide availability of these covariates. Certain jurisdictions may possess useful alternate data sources, such as emergency medical service (EMS) calls, insurance claims data, and measures for a mix of linked administrative datasets[Bharel et al., 2020], necessitating additional covariate consideration for enhanced model performance.

3.4.2 Conclusion

In an effort to better predict future fatal opioid-related overdose spikes and inform future harm-reducing interventions, we compared overdose forecasting options. Our study reinforces the value of intervention-aware metrics like %BPR in evaluating models for opioid overdose mortality forecasting. Our study also suggests that simple baselines like (weighted) historical averages should be included in future analyses, as more sophisticated and expensive-to-train models may not substantially outperform these baselines. As the opioid crisis continues to evolve, we hope our findings and our open-source resources enable improved model comparisons and better data-informed public health interventions that ultimately reduce the harm caused by overdose events.

3.5 Acknowledgments

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Chapter 4

Decision-aware Training of Spatiotemporal Forecasting Models to Select a Top-K Subset of Sites for Intervention

4.1 Introduction

Statistical machine learning methods for spatiotemporal forecasting can play a vital role in high-stakes applications from mitigating overdoses in public health [Marks et al., 2021c] to forest fire management [Cheng and Wang, 2008] to wildlife monitoring [Golden et al., 2022, Hefley et al., 2017]. Across these domains, there is a pressing need for predictive models that can make accurate predictions of near-term future events at fine spatiotemporal resolutions. Such models can enable inform data-driven decisions about how to allocate limited resources to maximize utility.

In this work, we seek to help decision-makers select *where to intervene*. Given historical data for a fixed set of S candidate spatial sites, we develop models that can recommend a specific subset of given size K for some action or intervention. We think of hyperparameter K as setting the budget for interventions. In an ideal world, decision-makers could afford interventions in all S sites. However, when resource constraints allow only K sites to receive interventions, a decision selecting a specific K-of-S subset is required. While such decisions may often be heuristic in current practice, we hope to offer data-driven solutions.

With this goal in mind, choosing a sensible performance metric is critical to assessing which models have real-world utility. Common metrics such as squared error or absolute error are not well matched to where-to-intervene decisions because they treat all sites equally. Recent work on overdose forecasting has suggested a metric termed the *fraction of best possible reach*, or *BPR* [Heuton et al., 2022, 2024]. BPR measures a ratio of event counts. The numerator sums over the model's recommended K sites, while the denominator sums over the best possible K selected in hindsight. This type of evaluation has been used in a preregistered trial [Marshall et al., 2022] for assessing forecasts of opioid overdoses in Rhode Island, as well as a follow-up feasability study [Allen et al., 2023]. BPR is applicable to many where-to-intervene problems beyond public health.

While some publications have reported BPR in evaluations, we suggest that a natural goal would be for this performance metric to inform two other key parts of data-driven decision-making: model-based *ranking* and model *training*. By ranking, we mean that given fixed model parameters, the knowledge of BPR as the metric of interest should impact the numerical score assigned to each site to determine the top K. By training, we mean how to update model parameters to achieve high BPR. This paper contributes new methods for solving both ranking and training problems when BPR is the preferred metric.

Our work overcomes several technical barriers. The first barrier is in ranking. Given a fixed probabilistic model, determining how to compute a per-site score, which will later be sorted to find the top K, is not obvious. It may be tempting to use the model's per-site mean, but decision theory suggests not all loss functions recommend the per-site mean as the best estimator [Berger, 2013, Murphy, 2022]. As a relatively new metric, the problem of how to assign a numerical ranking to sites for optimal BPR decision-making is currently open. We contribute a tractable ranking method that is provably best for a reasonable bound on BPR. We further show how the *score function estimator* [Kleijnen and Rubinstein, 1996, Mohamed et al., 2020] can be used to calculate gradients of this ranking with respect to parameters.

The second barrier prevents training parameters to improve BPR. First-order gradient descent is a common, effective algorithm we would like to use. However, gradients of BPR with respect to parameters are problematic. While small changes

Code: https://github.com/tufts-ml/decision-aware-topk to parameters induce some changes to per-site scores, only changes large enough to move a site into or out of the top-K ranked sites will adjust BPR. Thus, gradients of BPR with respect to parameters will be zero almost everywhere, preventing gradient methods from ever moving beyond subpar initial parameters. To fix this, we leverage recent advances in *perturbed optimizers* [Abernethy et al., 2016, Berthet et al., 2020] to yield effective and efficient gradient estimation for BPR. Our team explored this idea for optimizing BPR alone in earlier non-archival work [Heuton et al., 2023]; this paper offers an expanded treatment with more accessible presentation, while addressing two more barriers.

The final barrier is designing a training objective to achieve applied goals. We find that optimizing BPR alone can lead to predictions with far lower likelihood than conventional training. This raises concerns about overall model quality and generalization. To address this, we pose a constrained optimization problem to maximize likelihood subject to a BPR quality constraint. This combined objective delivers quality top-K recommendations *and* good forecasts for all sites. Our objective is reminiscent of past additive combinations of a regression loss and decision loss [Kao et al., 2009]. Unlike that work, ours pursues non-convex losses and directly enforces decision quality via constraints.

We ultimately contribute methods for how-to-rank and how-to-train when making where-to-intervene decisions. Using these tools, a variety of models can be directly optimized to make effective top K site recommendations. We demonstrate these contributions first on synthetic data, where we reveal how off-the-shelf methods without our innovations can be suboptimal for decision-making. We further evaluate against alternatives on two applications: mitigating opioid-related fatal overdoses and monitoring endangered birds. We hope our contributions spark interest in where-to-intervene problems in the methodological community and also lead to effective deployments of data-driven top-K decision-making in public health and beyond.

Related work. The application of machine learning to decision-making problems is widely studied in operations research literature [Bertsimas and Kallus, 2020,

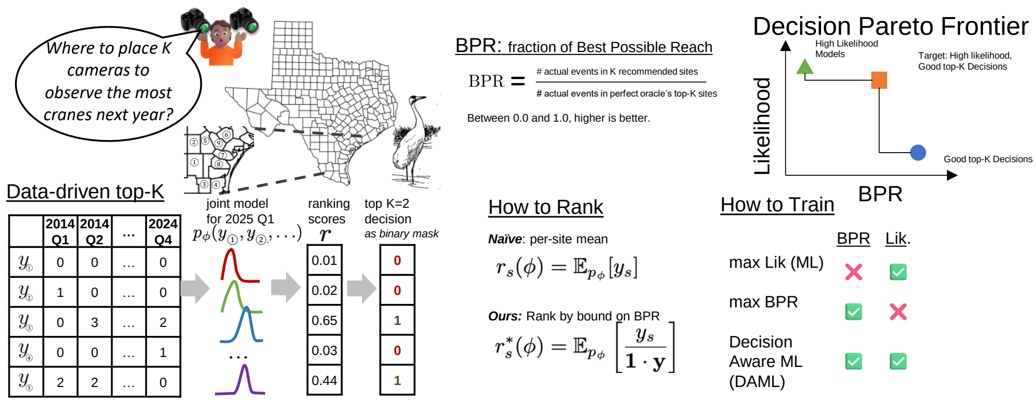


Figure 4.1: Visual overview of our approach and contributions to the *how to rank* and *how to train* open problems.

Sadana et al., 2025], including the problem of how to train a model for downstream decision-making [Mandi et al., 2024]. We review several model training approaches later in Sec. 4.4.

Other researchers have used decision-aware objectives to solve limited resource allocation problems. Chung et al. [2022] study how to allocate essential medicines in Sierra Leone across hospital sites. Gupta et al. [2024] pursue a where-to-intervene task in urban planning, selecting where to build speed humps to reduce pedestrian injuries. Our work differs in its focus on the BPR metric, our hybrid decision-aware objective that preserves likelihood, and evaluations that forecast the *future* given the recent past.

Throughout this paper, a recurring takeaway is that conventional training based on maximizing likelihood can yield suboptimal decision-making for BPR, especially when forecasting models are misspecified. In this vein, our work shares similar goals as *direct loss minimization* [Wei et al., 2021] and *loss-calibrated* methods [Lacoste–Julien et al., 2011]. We are inspired by the way these works combine task-specific losses and decision theory to improve probabilistic models. Others have extended loss-calibration to neural nets [Cobb et al., 2018] and to continuous actions [Kuśmierczyk et al., 2019]. Yet a tractable and scalable recipe that prioritizes top-K where-to-intervene decision-making is not an immediate next step from this work.

4.2 Technical Background and Problem Setup

Notation. Mathematically, some of our notation follows Sander et al. [2023]. Function TopKMask takes as input a vector $\boldsymbol{r}$ of length S and an integer K. It produces a binary vector of length S with exactly K entries equal to 1. Each 1 entry corresponds to a value in the top K largest entries of the input vector. The s-th entry of the output is

$$\textsc{TopKMask}(\boldsymbol{r}, K)_s = \begin{cases} 1 & \text{if } \textsc{rank}(\boldsymbol{r})_s \leq K \\ 0 & \text{otherwise} \end{cases}, \tag{4.1}$$

Here, function rank provides a numerical ranking (largest-to-smallest) from 1 to S for each entry of an S-dimensional input vector. The function TopKIDs acts on the same input as TopKMask, but return the size K set of integer indices corresponding to top K values.

Problem definition. We wish to probabilistically model events that occur across S distinct spatial sites over time. At each site, indexed by s, we can observe a *non-negative* scalar count or value $y_s \geq 0$. We assume that larger y_s corresponds to greater value in intervention at site s. In our public health applications, y_s represents counts of fatal opioid-related overdoses. In our later wildlife monitoring case study, y_s counts how often a rare animal appears. At each time t, we stack all observations into a vector $\boldsymbol{y}_t \in \mathbb{R}^S$. We assume that the true data-generating distribution for each vector $\boldsymbol{y}_t$ given past history does not change over time, including between the training and test periods.

We denote our joint model for this vector as $p_{\boldsymbol{\phi}}(\boldsymbol{y}_t)$, where model parameter vector $\boldsymbol{\phi}$ defines the density over r.v. $\boldsymbol{y}_t$. We assume that explicitly evaluating this pdf and sampling values of $\boldsymbol{y}_t$ are both feasible. Given these assumptions, our framework is quite flexible: the vector $\boldsymbol{\phi}$ could represent the weights of a neural network or the coefficients of a logistic regression or a Bayesian hierarchical model.

Given a training set of T times, our model family in general factorizes $p(\boldsymbol{y}_{1:T}) =$

$\prod_{t=1}^{T} p_\phi(\boldsymbol{y}_t|\boldsymbol{y}_{1:t-1})$. To ease notation throughout, we omit conditioning on past history or other exogenous features. So $p_\phi(\boldsymbol{y}_t)$ below should be read as equal to $p_\phi(\boldsymbol{y}_t|\boldsymbol{y}_{1:t-1})$ for models with such dependencies.

Our goal is to use this probabilistic model to solve a where-to-intervene decision making problem. We primarily intend to use the model to numerically rank all S sites, then select the top K sites in this ranking for near-future intervention.

Definition of BPR. At current time t, we evaluate a model ϕ's ability to select a top-K subset of sites for intervention before time $t+1$. Let $\mathcal{R}$ be the model's recommended subset of K sites among all S sites. For the rest of this section, let $\boldsymbol{y} \in \mathbb{R}^S$ be the vector of observations at the target time $t+1$ (we skip time subscripts on $\boldsymbol{y}$ to keep notation simple). The vector $\boldsymbol{y}$ is not available when the decision of $\mathcal{R}$ is made. Following Heuton et al. [2022], we define BPR as:

$$\mathrm{BPR}(\mathcal{R}, \boldsymbol{y}) = \frac{\sum_{s \in \mathcal{R}} y_s}{\sum_{s \in \textsc{TopKIDs}(\boldsymbol{y}, K)} y_s}. \tag{4.2}$$

Both terms in this fraction can only be evaluated in hindsight, after the vector $\boldsymbol{y}$ is realized at time $t+1$. The numerator counts how many events the model's recommendation would reach. The denominator counts how many events a perfect oracle with knowledge of the future could reach on the same budget of K sites. Overall, we interpret BPR as the fraction of events of interest the current model's selection $\mathcal{R}$ would reach compared to perfect knowledge of the future. Higher BPR indicates a better model for choosing where to intervene. BPR's best value is 1.0, its worst is 0.0.

Ranking sites. Given a fixed model ϕ and target time, the *ranking* problem is how to assign numerical values to all S sites so that if the top K sites are assigned to $\mathcal{R}$, we reap high BPR scores. We need to define a ranking vector $\boldsymbol{r} \in \mathbb{R}^S$ of numerical scalar scores for all S sites. Higher r_s values indicate greater priority for site s.

Suppose we have a loss function $L(\boldsymbol{r}, \boldsymbol{y})$ (not necessarily related to BPR) that produces a scalar value indicating the overall quality of taking an "action" $\boldsymbol{r}$ and then realizing outcome $\boldsymbol{y}$. Lower values of L indicate better decisions. A natural framework for making decisions about actions [Murphy, 2022] is to minimize the expected loss:

$$r^* = \arg\min_{\boldsymbol{r}} \mathbb{E}_{\boldsymbol{y}\sim p_\phi} \left[L(\boldsymbol{r}, \boldsymbol{y}) \right] \tag{4.3}$$

As a simplistic example, if the loss is defined as the sum of squared errors, $\mathcal{L}(\boldsymbol{r}, \boldsymbol{y}) = \sum_{s=1}^{S} (r_s - y_s)^2$, the optimal action is provably the per-site mean: $r_s = \mathbb{E}_{p_\phi}[y_s]$. Similarly, for the loss that sums up *absolute* errors, the optimal action is the per-site *median* [Balkus, 2024, Schwertman et al., 1990]. We tackle defining an optimal ranking for BPR.

To pose our ranking problem formally, we need to convert the higher-is-better BPR metric into a lower-is-better loss that depends on $\boldsymbol{r}$. Define *negative* BPR loss as

$$L^{\text{BPR}}(\boldsymbol{r}, \boldsymbol{y}) = -\frac{\boldsymbol{y} \cdot \textsc{TopKMask}(\boldsymbol{r}, K)}{\boldsymbol{y} \cdot \textsc{TopKMask}(\boldsymbol{y}, K)}. \tag{4.4}$$

This way of writing the loss with dot products of top-K binary vectors is equivalent to $-\text{BPR}(\textsc{TopKIDs}(\boldsymbol{r}, K), \boldsymbol{y})$.

Connection to 0-1 knapsack. Given a fixed $\boldsymbol{y}$ vector, the problem of selecting K sites to minimize L^{BPR} can reduce to the canonical 0-1 knapsack problem Dantzig [1957] where each site s would have value y_s and weight 1, and the budget constraint allows just K of all S sites. Our how-to-rank contribution solves a more general problem: how to set $\boldsymbol{r}$ when $\boldsymbol{y}$ is not given but must be forecasted by our model.

In Sec. 4.3 below, we show how analysis of tractable bounds of the loss in Eq. (4.4) suggests a high-quality ranking function $\boldsymbol{r}^*(\phi)$ for BPR. This ranking is usable across different model families, as long as the model p_ϕ allows generating many samples of events $\boldsymbol{y}$. Later in Sec. 4.4, we show how to *train* parameters ϕ with gradient descent to yield better top-K decisions as judged by BPR.

4.3 Methods for Ranking

4.3.0.0.1 Loose bound justifies per-site mean ranking. A natural first guess for ranking is the per-site mean: $\bar{\boldsymbol{r}} = \mathbb{E}_{p_\phi}[\boldsymbol{y}]$. We can show this is justifiable way to minimize expected loss on a simplistic upper bound on BPR. Assume there exists an *upper limit* U such that for all s, we can guarantee $U \geq y_s$. The sum over any K entries of vector $\boldsymbol{y}$ in the denominator of BPR is then bounded by $K \cdot U$. Plugging this bound into the minimize expected loss problem and simplifying with linearity of expectations yields

$$\boldsymbol{r}^* \leftarrow \arg\min_{\boldsymbol{r}} \underbrace{- \frac{\mathbb{E}_{p_\phi}[\boldsymbol{y}]}{K \cdot U} \cdot \textsc{TopKMask}(\boldsymbol{r}, K)}_{\leq \text{BPR}(\boldsymbol{r}, \boldsymbol{y})} \tag{4.5}$$

Many solutions exist: any vector $\boldsymbol{r}^*$ that satisfies $\textsc{TopKMask}(\boldsymbol{r}^*, K) = \textsc{TopKMask}(\mathbb{E}_{p_\phi}[\boldsymbol{y}], K)$ can be an argmin. One valid solution here is the per-site mean $\bar{\boldsymbol{r}}(\phi) = \mathbb{E}_{p_\phi}[\boldsymbol{y}]$. However, this solution is optimal for a potentially quite loose bound on BPR that approximates the denominator with the constant $K \cdot U$.

Tighter bound suggests the ratio estimator for ranking. Instead of bounding with constant $K \cdot U$, we can upper bound the denominator in Eq. (4.4) by summing over all S terms instead of the top K: $\sum_{s \in \textsc{TopKIDs}(\boldsymbol{y})} y_s \leq \sum_{s=1}^{S} y_s = \mathbf{1} \cdot \boldsymbol{y}$. This bound is tighter when the sum of all entries is less than $K \cdot U$, which is typically true of sparse $\boldsymbol{y}$ vectors in our applications. Our how-to-rank problem then becomes

$$\arg\min_{\boldsymbol{r}} \underbrace{- \mathbb{E}_{\boldsymbol{y} \sim p_\phi} \left[\frac{\boldsymbol{y}}{\mathbf{1} \cdot \boldsymbol{y}} \right] \circ \textsc{TopKMask}(\boldsymbol{r}, K)}_{\leq \text{BPR}(\boldsymbol{r}, \boldsymbol{y})} \tag{4.6}$$

Again, we solve via any vector $\boldsymbol{r}^*$ whose top-K binary vector $\textsc{TopKMask}(\boldsymbol{r}^*, K)$ equals $\textsc{TopKMask}(\mathbb{E}_{p_\phi}[\frac{\boldsymbol{y}}{\mathbf{1} \cdot \boldsymbol{y}}], K)$. One valid solution is the expected ratio of vector $\boldsymbol{y}$ to its sum, which we nickname the *ratio* estimator

$$\boldsymbol{r}^*(\phi) = \mathbb{E}_{\boldsymbol{y} \sim p_\phi} \left[\frac{\boldsymbol{y}}{\mathbf{1} \cdot \boldsymbol{y}} \right] \approx \frac{1}{M} \sum_{m=1}^{M} \frac{\boldsymbol{y}^{(m)}}{\mathbf{1} \cdot \boldsymbol{y}^{(m)}}. \tag{4.7}$$

This ranking is *distinct* from the per-site mean: there exist fixed models ϕ where the ratio estimator and the per-site mean would select different subsets of the same S sites. See App. 7.2.2 for concrete cases where the ratio estimator earns BPR 2.5x to 5x higher than the per-site mean, even when all estimators know the true data-generating model.

When exact computation of this expectation is not easy, we recommend a Monte Carlo approximation using M samples $\{\boldsymbol{y}^{(m)}\}_{m=1}^{M}$ drawn iid from p_ϕ, as in Eq. (4.7). This is a stochastic estimator; rankings can differ across repeat trials if M is not large enough.

4.4 Methods for Training

We now consider various ways to train the parameters of our probabilistic model on a training set that covers T distinct time periods indexed by t.

4.4.1 Maximum likelihood (ML) estimation

Conventional training would maximize the likelihood, or equivalently minimize negative log likelihood (NLL):

$$\mathcal{J}^{\mathrm{NLL}}(\phi) = -\textstyle\sum_{t=1}^{T} \log p_\phi(\boldsymbol{y}_t). \tag{4.8}$$

We assume p_ϕ is differentiable, so solving for a point estimate ϕ is possible via gradient descent. If we add an optional prior term $\log p(\phi)$ to enforce an inductive bias or control over-fitting, this is known as *MAP* estimation.

If the model is *well-specified* and training set size T is large enough, this is a reliable strategy to estimate ϕ. We could then use the ranking methods from Sec. 4.3 for where-to-intervene decisions. However, popular wisdom reminds us that "all models are wrong" in some way for real-world data. As we will show in later experiments, fitting a *misspecified* model via ML estimation can produce ϕ that deliver suboptimal BPR, even using the optimal ranking for that ϕ.

4.4.2 Direct loss minimization for BPR

Inspired by the broad goal of *direct loss minimization* [Wei et al., 2021], another approach would be to find parameters that minimize our BPR-specific decision making loss L^{BPR}. In this strategy, we seek ϕ values that minimize

$$\mathcal{J}^{\text{BPR}}(\phi) = \sum_{t=1}^{T} L^{\text{BPR}}(\boldsymbol{r}_t^*(\phi), \boldsymbol{y}_t), \tag{4.9}$$

Here, for $\boldsymbol{r}^*(\phi)$ we use the ratio estimator in Eq. (4.7).

To train with modern gradient methods, we'd need to compute the gradient $\nabla_\phi \mathcal{J} = \sum_t \nabla_\phi \boldsymbol{r}_t \cdot \nabla_{r_t} L_t$. However, technical difficulties arise with each term in this chain rule expansion. Below, we propose practical estimators for each term that overcome these difficulties.

Gradient $\nabla_\phi \boldsymbol{r}_t$. The difficulty here is differentiating through the expectation in Eq. (4.7), especially when $\boldsymbol{y}$ is a discrete random variable (integer counts in our later overdose or wildlife applications). We use the *score function trick* [Mohamed et al., 2020], popularized by Ranganath et al. [2014] yet dating back decades [Kleijnen and Rubinstein, 1996], sometimes also called REINFORCE [Williams, 1992]. We can draw M samples $\boldsymbol{y}_t^{(m)} \sim p_\phi$, then compute

$$\nabla_\phi \boldsymbol{r}_t = \frac{1}{M} \sum_{m=1}^{M} \frac{\boldsymbol{y}_t^{(m)}}{\mathbf{1} \cdot \boldsymbol{y}_t^{(m)}} \nabla_\phi \log p_\phi(\boldsymbol{y}_t). \tag{4.10}$$

This estimator can reuse the M i.i.d. samples already used to evaluate $\boldsymbol{r}_t$ in a forward pass. This is easy to implement for any model p_ϕ where sampling and evaluating the pdf is feasible, as we have assumed. We use automatic differentiation to compute $\nabla_\phi \log p_\phi(\boldsymbol{y}_t)$.

A downside of this estimator is high variance. We mitigate this with large M values, though future work may use *control variates* [Mohamed et al., 2020, Ranganath et al., 2014] or try other estimators for gradients of discrete expectations [Dimitriev and Zhou, 2021, Maddison et al., 2017].

Gradient $\nabla_{r_t} L_t$. For losses L defined in terms of TopKMask binary vectors, like BPR, it is difficult to compute useful gradients because this loss is flat almost everywhere with respect to the input rankings $\boldsymbol{r}$. To overcome this barrier, we leverage recent advances in *perturbed optimization* [Berthet et al., 2020], also referred to as *stochastic smoothing* [Abernethy et al., 2016]. A recent computer vision method [Cordonnier et al., 2021] shows how these ideas enable selecting a top-K set of patches from a high-resolution image for downstream prediction. We adapt this top-K approach to spatiotemporal forecasting for intervention.

Concretely, Cordonnier et al. [2021] obtain tractable J-sample Monte Carlo estimates of both the top-K indicator vector $\boldsymbol{b} = \textsc{TopKMask}(\boldsymbol{r}, K)$ and the Jacobian $\nabla_{\boldsymbol{r}} \boldsymbol{b}$ needed for backpropagation. First, we draw J independent samples of a standard Gaussian noise vector of size S: $\boldsymbol{z}_j \sim \mathcal{N}(0, I_S)$. Then, we compute

$$\begin{aligned} \hat{\boldsymbol{b}} &= \tfrac{1}{J} \textstyle\sum_{j=1}^{J} \boldsymbol{b}_j(\boldsymbol{r}), \quad \boldsymbol{b}_j(\boldsymbol{r}) = \textsc{TopKMask}(\boldsymbol{r} + \sigma \boldsymbol{z}_j). \\ \nabla_r \hat{\boldsymbol{b}} &= \tfrac{1}{J\sigma} \textstyle\sum_{j=1}^{J} \textsc{outer}(\boldsymbol{b}_j(\boldsymbol{r}), \boldsymbol{z}_j). \end{aligned} \tag{4.11}$$

The Jacobian $\nabla_{\boldsymbol{r}} \hat{\boldsymbol{b}}$ is an $S \times S$ matrix, where entry j, k gives the scalar derivative $\frac{\partial \boldsymbol{b}_j}{\partial r_k}$. The conceptual justification for the Jacobian estimator comes from Abernethy et al. [2016] and Berthet et al. [2020]. Noise level $\sigma > 0$ is a hyperparameter that sets the strength of stochastic smoothing. It must be carefully selected in practice to add enough noise so that indicators $\boldsymbol{b}_j$ change for different samples z_j, but not too much noise so the $\boldsymbol{b}_j$ preserve the signal in $\boldsymbol{r}$.

Putting our score-function trick and perturbed optimizer estimators together, we compute the overall gradient of loss at index t as a product of individual estimators: $\nabla_\phi L_t = \nabla_\phi \boldsymbol{r}_t \nabla_{\boldsymbol{r}_t} \boldsymbol{b}_t \nabla_{\boldsymbol{b}_t} L_t$. We compute the last term $\nabla_{\boldsymbol{b}_t} L_t$ via automatic differentiation.

Armed with this gradient estimator, we can pursue direct minimization of $\mathcal{J}^{\text{BPR}}$ via stochastic gradient descent methods. Stochasticity here comes from both M score function samples and J perturbation samples. For convenience and reliability, we use all T records in the training set in every estimate, avoiding minibatching over time.

We assume the true data-generating distribution is unchanged across train and test time periods. If this does not hold, objectives that just average over t as in Eq. (4.9) may have disadvantages. Instead we could upweight later t, or minimize out-of-sample error as in Gupta et al. [2024].

Other methods for decision-aware training. Our approach to direct BPR optimization here is an example of *decision-aware* or decision-focused training. In the taxonomy of Mandi et al. [2024], our approach is in the family of *differentiable perturbed optimizers*. Other work instead pursues *surrogate losses*. The *SPO+* method [Elmachtoub and Grigas, 2022] finds a convex surrogate for the "smart predict then optimize" optimization problem. The perturbed gradient (PG) method Huang and Gupta [2024b] develops a more sophisticated surrogate, with theory and experiments suggesting utility even with misspecified models. In both cases, surrogate bounds make SGD-based learning tractable for a wide set of optimization tasks, including our knapsack-like BPR problem but also other tasks like shortest path finding.

Downsides of only fitting BPR. When models are misspecified, directly estimating ϕ to minimize $\mathcal{J}^{\mathrm{BPR}}$ should yield better BPR than some ϕ' fit via conventional loss $\mathcal{J}^{\mathrm{NLL}}$, and better BPR means better decisions. However, probabilistic forecasts of near-future *outcomes* $\boldsymbol{y}_{t+1}$ produced by BPR-trained ϕ have questionable utility. Nothing in the $\mathcal{J}^{\mathrm{BPR}}$ objective makes p_ϕ accurately reconstruct even the train set $\boldsymbol{y}_{1:T}$; only *relative* ranking of sites matters. Even with high-quality decisions, a model which produces unlikely forecasts may be difficult to interpret or verify.

4.4.3 Decision-aware maximum likelihood

To jointly achieve the goals of good top-K decisions and accurate forecasts across all sites, we propose to find parameters that solve a constrained optimization problem:

$$\arg\min_{\phi} - \sum_{t=1}^{T} \log p_\phi(\boldsymbol{y}_t), \quad \text{s.t. } g_t(\phi) \leq 0 \ \forall \ t, \tag{4.12}$$

$$\text{where}\quad g_t(\phi) = \epsilon + L^{\mathrm{BPR}}(\boldsymbol{r}_t(\phi), \boldsymbol{y}_t).$$

Here, ϵ is a desired *lower bound* on a tolerable BPR for the decision task. Function $g_t(\phi)$ checks if the constraint is satisfied at time t, returning a non-positive value when $\mathrm{BPR}_t \geq \epsilon$ and a positive value otherwise. A feasible solution ϕ must deliver BPR as good or better than ϵ on the provided training set. Practitioners can set ϵ to achieve a desired minimum value for BPR. For example, if BPR below 60% was unworkable to stakeholders, set $\epsilon = 0.6$ to enforce $0.6 \leq \mathrm{BPR}$, recalling by definition $\mathrm{BPR} = -L^{\mathrm{BPR}}$.

We call this combined objective *decision-aware ML estimation*, or DAML. If the model is well-specified and training data are plentiful, DAML should deliver the same parameters as ML estimation when ϵ is low enough. However, when the model is misspecified and ϵ is higher than the BPR delivered by ML-estimated ϕ, we argue DAML's constraint will produce better top-K decisions than ML alone, trading lower likelihood for higher BPR. Compared to direct minimization of $\mathcal{J}^{\mathrm{BPR}}$, DAML can deliver similar BPR but more accurate forecasts of $\boldsymbol{y}$ for *all sites*. Additionally including the ML objective offers the ability to include an optional prior term $\log p(\phi)$ to incorporate any prior knowledge.

To solve in practice, we use the penalty method [Chong and Żak, 2013] to convert to an unconstrained loss:

$$\mathcal{J}^{\mathrm{DAML}}(\phi) = \textstyle\sum_{t=1}^{T} \lambda \max(g_t(\phi), 0) - \log p_\phi(\boldsymbol{y}_t). \tag{4.13}$$

This DAML formulation makes estimating ϕ via gradient descent possible. Here, $\lambda > 0$ is a nuisance hyperparameter that must be tuned. When the constraint is not satisfied, larger λ values force gradient updates to move parameters further in directions that might satisfy the constraint. In practice, we set λ such that both components of the loss are of similar magnitude during early training.

Implementation. Pseudocode for DAML training is provided in the supplement (Alg. 1). There are several key hyperparameters. First, M and J are the number of Monte Carlo samples used during training to estimate gradients via the score function trick and perturbed optimizer method. Setting M and J larger produces lower variance estimates, but at the cost of runtime and memory. Consequently, we recommend setting M and J as high as affordable. We found setting both to 100 worked for tasks in Sec. 4.5.

Proper selection of σ, the standard deviation of the Gaussian noise in the perturbed estimator, is also vital. If too small, estimated gradients will be zero; too large will swamp out any data-driven signal for learning. In practice, we found that σ values a bit smaller than the largest elements of rating $\boldsymbol{r}^*(\phi)$ worked well. Because our ratio estimator produces values between 0 and 1, we set σ between 10^{-3} and 10^{-1}.

4.5 Experiments and Results

Decision-aware training can benefit a variety of models in diverse problem domains. The subsections below cover toy and real applications. On each task, we fit models using all three training approaches from Sec. 4.4: ML estimation, BPR optimization, and DAML. We wish to verify common hypotheses throughout: (i) ML training can yield suboptimal top-K decisions; (ii) direct optimization of BPR improves this at the expense of likelihood; (iii) our DAML allow navigating tradeoffs between likelihood and BPR.

Common setup. In each task below, for a fixed task-specific K we first train models for ML and BPR. Using their final BPR values as guidelines, we select a suitable range of ϵ values for DAML and fit each one, intending to explore intermediate points on the two-objective Pareto frontier [Costa and Lourenço, 2015]. When training each objective, we run many random initializations to convergence across a range of learning rates and other hyperparameters (see App. 7.2.5.2 for details). We keep the model ϕ from one run that best achieved its objective on a validation set, early stopping as needed. A model's ultimate top-K rankings can have some stochasticity. Thus, later figures show *estimated distributions* of BPR across 1000

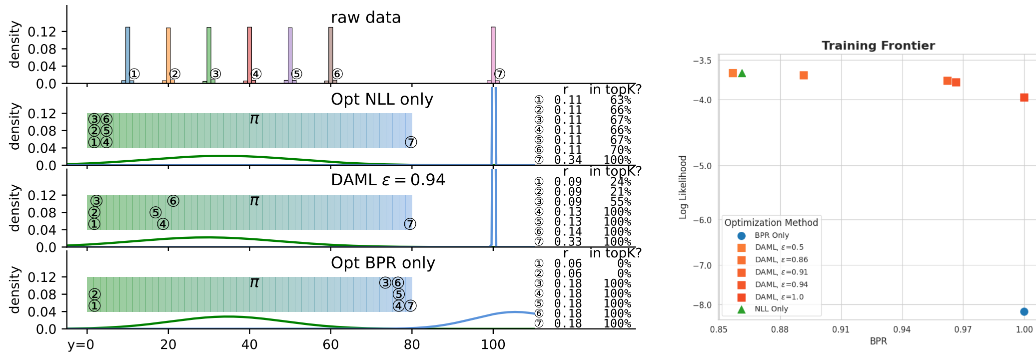


Figure 4.2: **Synthetic 1D data: learned models and Pareto frontier.** *Left Row 1:* histograms of y_s values by site (circled numbers). Sites 3-7 should be the top K=5 under the true model. *Left Rows 2-4:* Learned Gaussian components, with site-specific weights $\boldsymbol{\pi}_s$ marked as horizontal position between pure green and blue. Text provides ranking r with how often that site is in top $K = 5$ over 200 trials. *Right:* Likelihood vs. BPR tradeoff frontier for final models delivered by different training objectives.

trials of a M=1000 sample Monte Carlo estimate of $\boldsymbol{r}(\phi)$ after training is completed.

4.5.1 Synthetic Data

We begin with an illustrative synthetic case study chosen to highlight the trade-offs that decision-aware training allows a modeler to make when working with misspecified models.

Task. We create a synthetic dataset of $S = 7$ sites over $T = 500$ times. Integer data y_{ts} is generated i.i.d over time from a quantized Gaussian with site-specific mean and small constant variance. The first six sites have means evenly spaced between 10 and 60; the last site has a large mean of 100. Fig. 4.2 (top left) shows the training data.

Model. To show the benefits of our framework, we focus on a *misspecified model*. In particular, we model each site with a L-component positive Gaussian mixture model, with global mean and variance parameters and site-specific component frequencies:

$$p_\phi(y_s) = \sum_{\ell=1}^{L} \pi_{s,\ell} \cdot \mathcal{N}_+(y_s|\mu_\ell, \sigma_\ell^2) \tag{4.14}$$

Here $\phi = \{\mu_{1:L}, \sigma_{1:L}, \pi_{1:S,1:L}\}$. We reparameterize to unconstrained real values to make gradient-based learning possible: see App. 7.2.6 for details.

With L=7 components, the model would be well-specified and could recover the true data-generating process. However, we focus on misspecification, so we fit with L=2 components. This will hurt likelihood performance, as sites with distinct true means will need to use common means. However, we wish to show that solid top-K decision-making can still happen even with such severe misspecification.

Experiment setup. We use BPR with $K = 5$ for this task. For reproducible details, including all hyperparameters, see App. 7.2.6. We only report training set metrics here for simplicity; later tasks assess generalization to test data.

Results and analysis. From results in Fig. 4.2, we draw several conclusions. First, training to optimize NLL alone delivers subpar BPR for this task. Second, optimizing for BPR alone yields much better BPR values, suggesting that even this mispecified model can deliver much better top-K decision making than the off-the-shelf ML solution. However, BPR alone yields nonsensical likelihood values, as nothing in the objective forces ϕ to be good at modeling the outcomes y, only at relative ranking of the S sites. In Fig. 4.2, we see how our DAML hybrid objective allows a user to traverse the Pareto frontier of likelihood and BPR by enforcing a desired threshold on minimum BPR. As the desired minimum BPR threshold ϵ increases, we can sweep the tradeoff between likelihood and BPR. Ultimately, our DAML yields the best high-likelihood, high-BPR solutions in the top-right corner of the Pareto plot.

4.5.2 Opioid-related Overdose Forecasting

Motivation. The ongoing opioid overdose epidemic in the United States has incurred over 500,000 deaths in the past decade, with more than 80,000 fatal opioid-related overdoses in 2023 alone [Ahmad et al., 2025]. Possible evidence-based interventions to mitigate overdose fatalities include overdose education and nalaxone distribution. Scarce resources require local decision makers to allocate these interventions to *small areas* that are *high-risk* Allen et al. [2024], with co-incident

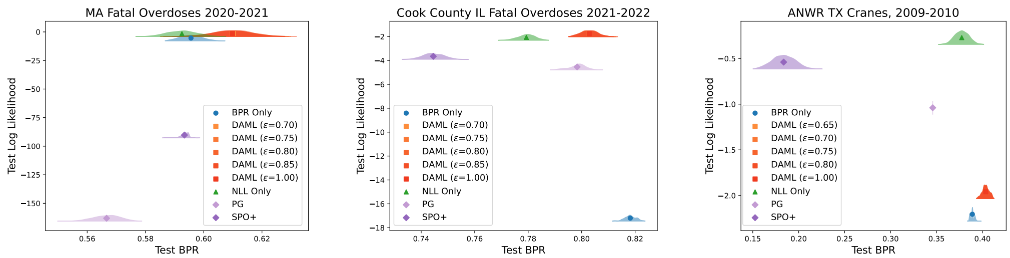


Figure 4.3: **Pareto frontier of best possible reach (BPR, x-axis) and log likelihood (LL, y-axis) for real-world tasks**. Higher is better on both axes. Each panel how the final models estimated by different training methods score on the test set of a forecasting task defined in Sec. 4.5. To capture the stochasticity of BPR due to our sampling-based ranking estimator, for each model we show an estimated density for BPR over 1000 trials. Uncertainty in this plot only corresponds to uncertainty in BPR, log likelihood is a point estimate. In all three tasks, our proposed decision-aware ML (DAML) delivers better top-K decisions as measured by BPR than ML estimation. DAML also delivers likelihood comparable to ML methods and *much better* than directly optimizing BPR. In the Cranes dataset, the DAML objective surprisingly offers better BPR than the BPR-only objective, although the magnitude of this difference is small and perhaps due to the small-scale and sparsity of this dataset.

education and support for proper follow-through. Public health agencies could use forecasting to help allocate limited resources towards the goal of harm reduction.

Several efforts have developed small-area forecasting models of opioid-related events [Bauer et al., 2023b, Marks et al., 2021c, Neill and Herlands, 2018]. A pre-registered trial for overdose reduction in Rhode Island [Marshall et al., 2022] used BPR-like metrics to evaluate the top-K predictions of conventionally-trained models. This past work does not *rank* or *train* to improve BPR, as we do.

Datasets. We study the capabilities of different methods on two datasets of historical opioid-related overdose mortality. Our IRB provided a Not Human Subjects Research determination for analysis of this decedent data. The first dataset, *MA Fatal Overdoses*, covers opioid-related overdose deaths in the state of Massachusetts from 2001-2021. This dataset is publicly available upon request from the MA Registry of Vital Records and Statistics. The second dataset, *Cook County IL Fatal Overdoses*, tracks opioid-involved overdose deaths in the greater Chicago area between 2015 and 2022. This is an open dataset obtained via the public website of the

Cook County Medical Examiner Case Archive [Cook County, IL, 2014-present].

We follow previous evaluations of these datasets in Heuton et al. [2024]. Each dataset was processed to a common format of fatal overdose counts per time and spatial unit. For the spatial units, we chose *census tracts*. Each tract by design contains a mean population of 4000 people, a scale that allows capturing variation in overdoses at the neighborhood level. For temporal binning, we picked calendar years to reflect the frequency at which health agencies might enact policy changes. Summary facts are in Tab. 7.1.

Task. Our forecasting task is to predict the next year's count of opioid-related fatal overdoses in each census tract. For *MA*'s S=1620 tracts, we train on data from 2011-2018, tune hyperparameters on validation data from 2019, and test on 2020 and 2021. For *Cook County IL*'s S=1328 tracts, we train on data from 2015-2019, tune on 2020, and test on 2021 and 2022. These splits follow Heuton et al. [2024].

Given a trained model, we assess heldout likelihood over all sites as well as BPR with K=100 to measure the model's where-to-intervene ranking. A K=100 budget was selected to reflect realistic public health budgets, and is similar to values used in other studies [Marshall et al., 2022].

Model. A recent benchmark [Heuton et al., 2024] compared many models designed for fatal overdose forecasting, including neural architectures with attention [Ertugrul et al., 2019b] and more classical statistical models. A top-performing model is *negative binomial mixed-effects regression* [Marks et al., 2021b]. The generative model can be expressed as

$$\begin{aligned} y_{st} &\sim \mathrm{NegBin}(\mu_{st}, q), \qquad (4.15)\\ \log(\mu_{st}) &= \beta_0 + \boldsymbol{\beta}^T \mathbf{x}_{st} + b_{0s} + b_{1s} t. \end{aligned}$$

Here, count y_{st} is modeled as a Negative Binomial, where the log of the number of successes to stop at μ_{st} is a linear function of feature vector $\mathbf{x}_{st}$ as well as a site-specific intercept b_{0s} and site-specific b_{1s} weight on time t. The parameter q is a probability of success shared by all sites.

Feature vector $\boldsymbol{x}_{st}$ includes tract s's *overdose gravity*, a recent average of opioid-related overdose deaths in tracts spatially near to s, as well as measures of sociological vulnerability. See App. 7.2.5.3 for details.

The overall parameters to estimate during training are $\phi = \{q, \beta_0, \beta, b_{0,1:S}, b_{1,1:S}\}$. To make constrained parameters amenable to gradient descent, we employ suitable one-to-one transforms to unconstrained spaces. Random effect weights $\boldsymbol{b}$ are regularized via a *prior* that assumes a zero-mean Normal distribution with learnable covariance parameters. Full details are provided in App. 7.2.7.

Competitor methods. We compare to two other decision-aware methods discussed above: Perturbed Gradient (PG) [Huang and Gupta, 2024b] and SPO+ [Elmachtoub and Grigas, 2022], as implemented in PyEPO software [Tang and Khalil, 2024b]. To be fair, each uses the same model p_ϕ, the ratio estimator to rank sites, and the gradient of this estimator in Eq. (4.10). We conduct a hyperparameter search over learning rate (and perturbation noise for PG), selecting the model with the best loss value on validation data.

Setup. We followed the common setup described above. For reproducible details specific to overdose tasks, see App. 7.2.5.2.

Results. Results on *test* data for both MA and Cook County IL tasks are shown in Pareto frontier plots in Fig. 4.3. For both datasets, we see that ML training yields suboptimal top-K decisions. Direct optimization of BPR can improve BPR, though gains on test data over ML vary (+0.04 on Cook County; less than +0.01 on MA). Direct BPR and surrogate loss (SPO+, PG) solutions can yield *much worse* likelihood, as expected. The surrogate loss methods provide lower quality decisions than the DAML and direct BPR approach. Our DAML approach provides better top-K decisions than the ML approach, with no visible decay in likelihood.

4.5.3 Endangered Bird Forecasting

Motivation. In the 1940s, whooping cranes were almost completely extinct in the U.S., with only 20 existing in the wild [Cannon, 1996]. Thanks to efforts over the years to preserve their population, there are roughly 650 wild cranes today. This key species is still listed as endangered in 2025.

A major flock of cranes, known as the Aransas-Wood population, winters at the Aransas National Wildlife Refuge (ANWR) along the Gulf Coast of Texas [Vartanian, 2023], while spending summers north in Canada. Ecologists wish to actively monitor this population. Regular aerial surveys of the Texas wintering region have been conducted for decades [Taylor et al., 2015]. Using binned spatiotemporal data of sighting counts over time from these surveys, we wish to offer data-driven forecasts of where cranes may be found. This could help conservationists decide where to send future human monitors or where to place K fixed-location cameras to efficiently track population health.

Data. We use raw data from Taylor et al. [2015], which digitized decades of aerial surveys of ANWR that marked individual crane sightings on paper maps. We processed this *ANWR TX Cranes* data into a common format of sighting counts over time and space, selecting bin sizes to support our goal of using the top-K sites to improve monitoring. Quality cameras or binoculars that could be used to track cranes can reasonably capture a 1.5 meter tall whooping crane in a 250-meter radius. Therefore, for spatial units, we divided the ANWR into 1338 boxes, each 500 meters per side. For temporal binning, we use 2 month periods. This choice catches seasonal variation, but avoids how finer scales might burden staff to move cameras too often.

Task. The experiment on these data was trained on years 2002-2006, validated on 2007-2008, and tested on 2009-2010. In this context, a "year" represents one wintering season; winter 2010 means the winter that began in October 2010 and ended in April 2011. We choose $K = 50$ for this task. This is an estimate of how many sites scientists might reasonably monitor every two months on a limited budget.

Models. The same negative binomial mixed-effect model was used as in Sec. 4.5.2. Features $\boldsymbol{x}_{st}$ for site s at time t include historical bird counts at s, the latitude and longitude of the site's centroid, time of year, and the overall time.

Setup. We followed the common setup described above. See reproducible details in App. 7.2.5.5.

Results. From results in Fig. 4.3 (right panel), we find that optimizing NLL or

BPR *alone* will yield poor results in the other metric. In this case, direct optimization of BPR results in the best top-K decisions, but dramatically worse likelihood. Our DAML models improve BPR by up to 0.05 over conventional ML training with only modest decay in likelihood. Our DAML and direct BPR objectives also deliver better BPR than previous decision-aware methods (PG, SPO+). This is even after providing smarter initializations to these methods, as we found their training had trouble improving on our common random initialization of ϕ, perhaps due to this task's much sparser $\boldsymbol{y}$ values.

4.6 Discussion

We have addressed two open challenges related to top-K resource allocation problems guided by the best possible reach (BPR) performance metric. We provided a ranking strategy that can outperform simple per-site means. We posed a training objective that strives for high likelihood across all sites while ensuring top-K decisions meet a stakeholder-specified quality level. Our experimental evaluations suggest our approach can better manage tradeoffs in likelihood and BPR than conventional training methods or previous decision-aware methods.

There are several limitations to this study. We focused on showing the tradeoffs between likelihood and BPR for a fixed model family in each task without comparing a wide variety of possible models. Only one type of model misspecification is considered each in the synthetic and real-world experiments; perhaps decision-aware objectives are more or less different to maximum likelihood estimates depending on the kind of model misspecification. Our evaluations use fixed K values and do not explore sensitivity to K or other hyperparameters. Practitioners may need multiple metrics to assess overall utility; focusing myopically on BPR may not always be wise. Additionally, our framework assumes any site with large outcome y_s is a better candidate for intervention. Future work could explore data-driven site selection that considers how site-specific attributes might make some interventions more or less effective.

Looking forward, we hope to see applications of these ideas inform public health, wildlife conservation, and other where-to-intervene decision-making problems across the private and public sectors. We also hope that future methodological work could scale up to much larger problems ($S > 2000, T > 20$) as we found that our current experiments tested the limits of commodity GPUs.

Chapter 5

Unlocking the Secrets of Perturbation Methods for End-to-End Prediction and Optimization

5.1 Introduction

Machine learning (ML) models may inform real-world decisions with high-stakes consequences. Examples include "Is air quality sufficient to go outside now?" [Berlinghieri et al., 2026], "Where should scarce public resources be allocated?" [Heuton et al., 2025], or "Which route should I take?"[Tang and Khalil, 2024a]. Ideally, prediction models would be directly trained to optimize a loss tailored to the stakes of the decisions at hand. In most cases, this *decision-aware* end-to-end training should make superior decisions [Elmachtoub et al., 2025a].

Yet many decision problems of practical interest have a loss that is piecewise constant as a function of model parameters. Consequently, gradients of the loss with respect to parameters will be 0 almost everywhere. All zero gradients mean that gradient descent, one of ML's go-to tools for parameter learning, will be ineffective.

To overcome this limitation, many methods have been proposed [Mandi et al., 2024]. We consider two families of methods here. In Perturbation methods [Abernethy et al., 2016, Berthet et al., 2020], informative gradients are obtained by adding noise to model outputs. The exemplar method we focus on here is the Differentiable Perturbed Optimizer (DPO) [Berthet et al., 2020]. In contrast, Surrogate Loss methods pick another less-flat loss with non-zero gradients instead of the ideal (yet flat) decision loss. Exemplars here include methods known as PG [Huang and Gupta, 2024a] and SPO+[Elmachtoub and Grigas, 2022]. Using their informative gradients, both families estimate parameters via the common toolbox of gradient descent.

Among the many methods now available, which is recommended for practical use? In recent years, several methods papers and benchmark studies have begun to offer apples-to-apples comparisons of different methods on synthetic and real problems. However, we find that Perturbation methods have rarely been featured in such experiments. Despite discussing Berthet et al.'s DPO in related work, at least two high-profile recent studies [Geng et al., 2024, Huang and Gupta, 2024a] omit DPO entirely from their experiments.

In this work, we argue that DPO is a competitive method worth considering. Our contributions are:

- We evaluate DPO alongside recent end-to-end learning methods across three experimental settings: Geng et al.'s benchmarks (Fig. 5.1), a synthetic dataset from Huang and Gupta (Fig. 5.2), and applications to top-K site selection (Fig. 5.3).
- While DPO can work well out of the box, we show several problems where DPO's success is highly sensitive to its noise-level hyperparameter. With improved tuning strategies beyond simple grid search, we find DPO can noticeably improve its regret, though on some tasks no big gains occur.
- We further reveal that a strong surrogate method well-suited for misspecified real-world scenarios, PG [Huang and Gupta, 2024a], also has a key hyperparameter sensitivity where even recommended settings can yield subpar regret (see Fig. 5.2).

Our work suggests that perturbation-based methods deserve greater consideration in decision-focused learning pipelines, particularly when surrogate losses are misaligned with true decision objectives. We hope this practitioners in many domains make better decisions via end-to-end learning.

5.2 Background: End-to-end Methods

In our problems, a decision $z \in \mathcal{Z}$ must be chosen from a discrete feasible set to minimize a cost that depends on outcomes Y that are unknown at decision time. We model Y as a random variable produced from a conditional distribution given observed features $X = x$. For example, in a later top-K site selection tasks each z

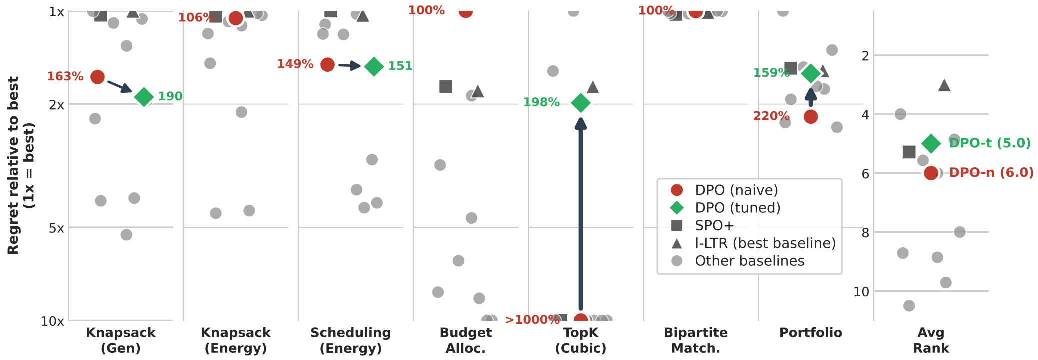
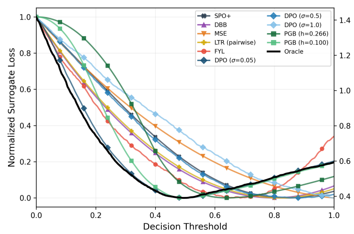


Figure 5.1: **7 task benchmark from Geng et al..** Each column is a task. DPO: naive grid search vs. extra tuning.



Method	Opt. thr.	Regret
$\mathrm{PGB}_{h=0.1}$	0.50	0.39
$\mathrm{DPO}_{\sigma=0.05}$	0.50	0.39
$\mathrm{PGB}_{h=0.266}$	0.64	0.44
$\mathrm{DPO}_{\sigma=0.25}$	0.68	0.45
FYL	0.69	0.46
DBB	0.80	0.51
LTR (pairwise)	0.82	0.52
SPO+	0.85	0.53
$\mathrm{DPO}_{\sigma=0.5}$	0.89	0.55
MSE (2-stage)	1.0	0.59

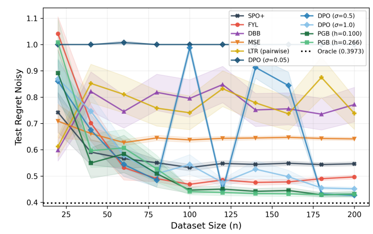


Figure 5.2: **Results on synthetic data under misspecification for various end-to-end PnO methods.** Left: Loss as a function of decision threshold. Ideal methods should match the oracle (solid black), esp. in minima location. Center: Regret for each method at its optimal θ. Right: Regret vs. sample size n, showing effectiveness of gradient descent for θ.

represents an indicator vector that picks K sites for intervention out of many, while vector y has an observed score for each site, with smaller values indicating desirable sites given x. The decision to do z when the outcome is y incurs a scalar cost $c(z, y)$.

The decision-making problem is, given an x, to solve for the value(s) of z with lowest expected cost,

$$z^\star(x,\theta) \;:=\; \arg\min_{z\in\mathcal{Z}} \mathbb{E}_{p_\theta(Y|X=x)}[c(z,Y)]\,. \tag{5.1}$$

Here, the parameter $\theta \in \mathbb{R}^P$ defines a conditional distribution over Y given $X{=}x$. At decision-making time ("test time"), only features x and parameter θ are available; the realization of Y is unknown until later.

The training problem is to estimate θ given observed data $\{(x_i, y_i)\}_{i=1}^n$ assumed drawn i.i.d. from the joint distribution of (X, Y). We could learn θ to maximize the likelihood $p_\theta(Y|X)$ given observed data. However, end-to-end methods seek an estimated θ that delivers good decisions via Eq. (5.1) when used on never-before-seen contexts $X{=}x'$ drawn from the same distribution. To solve this, training can be framed as a cost minimization problem: $\theta^* \leftarrow \mathrm{argmin}_\theta \sum_{i=1}^n c(z^*(x_i, \theta), y_i)$.

Linear cost assumption. A common assumption [Huang and Gupta, 2024a] is that cost is a linear inner product $c(z, y) = \langle z, y \rangle$. Here z and y are vectors of size D and decisions simplify due to linearity of expectation: $z^*(x, \theta) = \mathrm{argmin}_{z \in \mathcal{Z}} \langle z, \hat{y}_\theta(x) \rangle$, where $\hat{y}_\theta(x)$ denotes the mean of Y under $p_\theta(Y|X = x)$. Even here, the decision problem is piecewise constant as a function of $\hat{y}_\theta$. Its flatness yields all-zero gradients which break gradient-based learning of θ.

DPO method. The differentiable perturbed optimizer [Berthet et al., 2020], addresses this issue by smoothing the decision map via random perturbations of the predictions. For noise ξ from a standard Normal or Gumbel distribution with noise-level $\sigma > 0$, the perturbed decision is defined as

$$z_\sigma(x, \theta) \;\coloneqq\; \mathbb{E}_\xi\big[\; \arg\min_{z \in \mathcal{Z}} \langle z,\, \hat{y}_\theta(x) + \sigma \xi \rangle \;\big]. \tag{5.2}$$

In practice, we can estimate this perturbed decision vector of size D and its gradient via Monte Carlo estimates. We focus on iid Gaussian noise and assume M samples, each one denoted $\xi^{(m)} \sim \mathcal{N}(0, I_D)$. Define the decision obtained with each noise sample as as $\hat{z}^{(m)} {=} \arg\min_{z \in \mathcal{Z}} \langle z,\, \hat{y}_\theta(x) {+} \sigma \xi^{(m)} \rangle$, then we can estimate the perturbed decision $\hat{z}$ and the Jacobian matrix J needed for gradients as

$$\begin{aligned} \hat{z}(x, \theta) &= \tfrac{1}{M} \textstyle\sum_{m=1}^M \hat{z}^{(m)} \\ \hat{J}(x, \theta) &= \nabla_\theta \hat{y}(x, \theta) \Big[\tfrac{1}{\sigma M} \textstyle\sum_{m=1}^M \textsc{outer}(\hat{z}^{(m)}, \xi^{(m)}) \Big] \end{aligned} \tag{5.3}$$

This Jacobian estimator $\hat{J} \approx \nabla_\theta z_\sigma(x, \theta)$, a $P \times D$ matrix, comes directly from the chain rule and Prop. 3.1 of Berthet et al. Noise distributions other than Gaussian require changes to the outer product term.

DPO hyperparameters. DPO's most sensitive hyperparameter is noise-level σ, which controls a bias–smoothness tradeoff. Smaller σ values yield high-variance or vanishing gradients. We recover noise-free decisions $z^*(x, \theta)$ as $\sigma \to 0^+$. Larger σ induce smoother decision maps at the cost of biasing decisions away from $z^\star$.

As typical in Monte Carlo methods, DPO's estimators become more accurate as the number of samples M increases: larger M produce lower-variance estimates at the cost of an increased number of calls to the (often expensive) optimization solver that produces $\hat{z}^{(m)}$.

5.3 Experiment A: Synthetic Task

We pursue a synthetic task from Huang and Gupta (details in App. 7.3.1), with univariate x, y, and z. The true data-generating process for y given x relationship is piecewise linear, as shown in Fig. 7.1. The mean of y is above zero when $x < 0.5$ and below otherwise. Asymmetric noise is added to this mean. The decision task here is to select a binary z value ($\mathcal{Z}$=$\{0,1\}$) to minimize cost $z \cdot y$. The optimal decision strategy is thus to guess $z = 0$ when $x < 0.5$ and $z = 1$ otherwise.

Model. We use a 2-parameter linear model: $\hat{y}_\theta(x) = \theta_1 x + \theta_0$. This is misspecified to the true "hockey stick" trend, but it can still yield optimal decisions. As long as the learned zero-crossing occurs at $x = 0.5$, there are many possible θ which would achieve optimal cost.

Results. In Fig. 5.2 we replicate the experiments of Huang and Gupta [2024a], adding DPO at various σ. In the left panel, we explore whether each method's loss is suitably aligned to the ideal. here we vary the location of the decision threshold implied by the parameter θ and calculate each method's loss function at that θ on a test set of 10000.

We highlight two findings. First, with an appropriate noise level σ, DPO aligns with the true decision loss as well as PG, even under model misspecification. DPO was cited in Huang and Gupta, but not included in experiments. Second, PG is critically dependent on its finite-difference hyperparameter h; small increases can make PG's surrogate loss no longer have a minima that coincides with the optimal. PG does well with h=0.1, but h=0.266 as recommended in Huang and Gupta's code is subpar.

In Fig. 5.2's right panel, we examine methods for the quality of estimated θ from gradient descent (GD) as train size n increases. We plot cost on the test set, showing uncertainty over 100 trials (separate datasets and runs of GD). These plots show that even though DPO's loss itself is more aligned with true decision loss at low σ, higher σ may produce better θ via GD.

5.4 Experiment B: PnO Benchmark

We replicate 7 tasks from Geng et al. [2024]'s end-to-end prediction and optimization (PnO) benchmark, using their implementation and settings. They use synthetic and real datasets from energy, citation networks, and beyond for a variety of decision tasks. See App. 7.3.2 for task specific descriptions.

We add a DPO method to the benchmark based on J. Tuero [2021]'s code. While Geng et al.'s provided software contains some DPO code, it is not fully implemented. Results for the 11 baseline methods are taken from the original benchmark.

Training DPO. Our basic or "naive" DPO completes a grid search of noise-level $\sigma \in [0.1, 0.5, 1]$ and learning rate $[0.01, 0.005, 0.001]$, selecting the lowest-cost run on the benchmark's validation set. These 9 runs for DPO may advantage it compared to other methods (which received 5 different learning rates and no tuning of other hyperparameters). Yet our goal is to highlight what is possible with DPO, not claim superiority.

Results. Figure 5.1 shows DPO's performance across the 7 tasks and 11 methods studied by Geng et al.. DPO achieves the best performance on bipartite matching and budget allocation, near-optimal performance on the real-world knapsack energy problem, and ranks in the middle for 2 more tasks.

Here, some results contrast with the findings of Geng et al. [2024]. While the authors say "methods in the discrete category [such as DPO] do not perform as well as others, especially on non-linear optimization tasks (budget allocation)," we find that DPO with basic grid search performs the best at the Budget Alloc. task.

Extra tuning. If, after the initial size-9 grid search, DPO was in the bottom half of the methods compared, we performed further tuning to illustrate what can help and harm DPO's performance.

On the TopK task where DPO-basic scored quite poorly, our extra tuning involved decaying σ linearly to 0 over iterations, recommended by Cordonnier et al. [2021]. We think this decay is helpful due to the nature of the problem: a decision boundary which rarely appears in the training data matters a lot in overall test cost. For this task, the model $\hat{y}_\theta$ is well-specified, capable of perfectly recovering the data generating process. Thus, a two-stage rather than end-to-end method performs best overall, which is explained by proofs in Elmachtoub et al. [2025b].

The energy task exposes one of the major weaknesses of DPO: each noisy sample requires its own solver call, which is expensive here. As such, we were only able to use M=1 sample when training on this task, and we believe this to be the reason for its failure. Further tuning (more σ values, σ decay) were unsuccessful: we obtained better regret on the validation set, but slightly worse test-set performance.

Similarly, it was challenging to improve performance on the synthetic Knapsack task. Because σ is either constant or decaying, it is important that the predicted outcome y stays in a relatively stable range of magnitudes. As such, we placed a tanh activation on the outputs of the model for this task. Again, we see this improves validation but harms test-set performance.

Finally, the portfolio task is a continuous quadratic problem, and therefore already has informative gradients and requires no alternative optimization objective. Accordingly, DPO-tuned in this problem corresponds to optimizing portfolio returns directly.

5.5 Experiment C: Top-K Site Selection

In Heuton et al. [2025], the authors study top-K site selection problems on real data from public health and wildlife conservation. They find that a method based on DPO performs well on a heldout decision-making task, while both SPO+ and PG

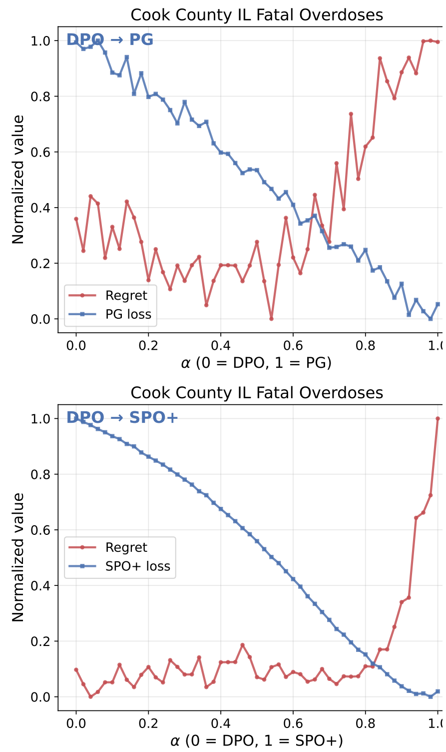


Figure 5.3: **Top-K site selection:** Comparing estimated θ from DPO and SPO+/PG. Expanded version in Fig. 7.2.

scores worse than DPO. Yet *why* DPO wins is unexplained in the original work. Does DPO achieve superior results because it has better agreement with decision loss? Or are PG and SPO+ methods stuck in local minima?

We attempt to answer these questions on all 3 tasks from Heuton et al., by interpolating between the parameters θ learned by DPO and the θ estimated by either PG or SPO+. Fig. 5.3 shows results for the overdose death prevention task in Illinois. Expanded results for all tasks are in Fig. 7.2. Across 2 methods and 3 tasks, the DPO estimated θ always has lower decision regret while almost always having higher surrogate loss (PG or SPO+). This means the **surrogate loss itself appears inadequate** in these tasks.

5.6 Conclusions

We hope this paper provides evidence that DPO can be successful, but also that tuning its hyperparameters remains challenging and other structural aspects (like reliability of gains on small validation sets generalizing to test) remain key barriers.

Our recommended procedure would be to first carefully tune the perturbation magnitude. If necessary, decay this parameter to zero to eliminate the gap between the training and evaluation objective. Finally, while we believe it is prudent to constrain the magnitude of the predicted outcomes, we found this did not provide a noticeable benefit on these benchmarks.

Chapter 6

Conclusion and Outlook

We can make better decisions everywhere

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Chapter 7

Supplementary Material

7.1 Supplementary Material for Chapter 3

7.1.1 Additional Model details

Model hyperparameters are selected using the year prior to the test years as a validation year: 2019 for Massachusetts and 2020 for Cook County. When validating, models are trained through the year prior to the validation year. For evaluation on the test years, models are retrained through the validation year. Hyperparameters are selected by selecting the model with the highest BPR. In Massachusetts, we consider using up to 10 years of historical data when training (2010-2019). In Cook County there are fewer years of available historical data, and so training is limited to 2016-2020.

7.1.1.1 All-Zeroes

The All-Zeroes model is presented to highlight two things: the BPR of a naive policy, and the RMSE and MAE of a naive model. This model is very simple: every prediction is always 0 fatal overdoses. However, this presents a challenge for calculating BPR: what are the top K locations if every location is tied? In this case, we take 10,000 samples, and randomly pick the K locations to serve as the numerator for BPR. We then calculate the BPR for each of 10,000 samples, and report the average.

7.1.1.2 Last Year

For this model, the prediction is simply the previous year's fatal overdose count. When predicting for the second evaluation year (2021 in Massachusetts and 2022 in Cook County), the fatal overdose count from the first evaluation year is used. This is subtly different from the behavior of the regression models, where the models are trained using no data from the evaluation years.

7.1.1.3 Historical Average

In this model, the output is an un-weighted average of historical mortality. For both Massachusetts and Cook County 4 years are selected. These are both selected via the validation year. When predicting for the second evaluation year (2021 in Massachusetts and 2022 in Cook County), the fatal overdose count from the first evaluation year is used. This is subtly different from the behavior of the regression models, where the models are trained using no data from the evaluation years.

7.1.1.4 Weighted Historical Average

This model is a linear regression on historical fatal overdose count using only past years mortality as a predictor. Scikit-learn's[Buitinck et al., 2013] ridge regression is used, which performs L2-regularized regression. The regularization strength α is selected via hyperparameter search by trying 29 evenly spaced values on a log scale between 10^{-6} and 10^{8}. For Massachusetts, 10 years of historical data and an α of $10^{4.5}$ is used. For Cook County, 3 years of historical data are used and an α of $10^{4.5}$ is selected.

7.1.1.5 Linear Poisson GLM

This uses Scikit-learn's[Buitinck et al., 2013] Generalized Linear Model with a Poisson Likelihood and a log link. The hyperparameters explored are the number of prior years of mortality to include in the model and the L2 regularization strength α. Up to 10 years of previous mortality were considered for Massachusetts and 6 years for Cook County. The model is run with and without social vulnerability covariates.

Without social vulnerability covariates: In Massachusetts 10 years of prior mortality are used in the model with an α of 1. In Cook County, 5 years of historical data are used with an α of $10^{0.5}$.

With social vulnerability covariates: In Massachusetts 6 years of prior mortality are used in the model with an α of 1. In Cook County, 5 years of historical data are used with an α of 10.

7.1.1.6 Gradient Boosted Trees

This uses Scikit-learn's[Buitinck et al., 2013] Histogram-basedGradient Boosted Trees model. We considered both squared-error and Poisson loss functions. We tested using both 32 and 128 maximum iterations. For the minimum samples per leaf we used 9 equally spaced values between 2^0 and 2^8 on a log-2 scale. The maximum number of leaf nodes tested were 5 equally spaced values between 2^4 and 2^8 on a log-2 scale. Up to 10 years of previous mortality were considered for Massachusetts and 6 years for Cook County.

7.1.1.7 Gaussian Process Models

Here we use Scikit-learn's[Buitinck et al., 2013] Gaussian Process (GP) implementation. Due to the high computational cost of GP models, and following prior work[Allen et al., 2023], we only consider up to 5 years of historical data for both Massachusetts and Cook County, and we omit social vulnerability covariates. As in the prior work,we use a kernel that additively combines a Radial-Basis Function (RBF) kernel with a white noise kernel. The initial length scale of the RBF kernel is set to 0.5, and the noise level bounds on the white noise are set to (10^{-5}, 10^1). The outcomes are normalized to 0-mean and unit variance. Up to 9 restarts of the optimizer are used.

7.1.1.8 CASTNet

While we attempted to follow the original implementation of CASTNet[Ertugrul et al., 2019a] as closely as possible, there are some significant modifications. The original CASTNet was concerned with predictions at a fine temporal-resolution, weekly. However, our initial experiments found little benefit at this scale, and we consider a much coarser scale: annual predictions. We lack the high-resolution crime data that the original CASTNet project uses as dynamic covariates. Furthermore, while we do have demographic and economic variables (the 5-dimensional Social Vulnerability index), these are not static at the annual scale but dynamic, accordingly these are used as the only dynamic covariates. For static covariates, only the latitude and longitude of the census tract are used. We use the hyperparameters selected by the original work: the LSTMs have a hidden unit size of 32 with a dropout value of 0.1, the group-level regularization coefficient is 0.0025, and the optimizer used was Adam with a learning rate of 0.5. Given that we have 2 evaluation years, we train the model twice, once with a lag time of 1-year and again with a lag time of 2-years. The 1-year lag model is used to predict for the first evaluation year (2020 in MA and 2021 in Cook County) and the 2-year lag model is used to predict for the second. This way no training data leaks into the model.

7.1.1.9 Bayesian Spatiotemporal Models

In the original work[Bauer et al., 2023a] on Bayesian Spatiotemporal models for opioid overdose forecasting, three separate models are proposed. All three models use an Autoregressive-1 term to model temporal dependence, and two of the models use spatial correlations. Because all three models are reported to behave similarly, we use what is called "Model 1", lacking spatial correlations. Furthermore, the authors state that any number of temporal terms could be used, but do not specify which. Here, we choose a linear temporal term.This model is implemented using R-INLA[Rue et al., 2009]. Linear coefficients are used when adding the social vulnerability covariates.

7.1.1.10 Negative Binomial Regression with Spatially Lagged Covariates

The authors of this method[Marks et al., 2021a] helpfully provide code to run this model which we were able to use with little modification. Census tract level population estimates are taken from the same survey data as the social vulnerability covariates. The carrying capacity is initialized to 5% of the population in the first year of training data (2010 for MA, 2015 for Cook County).

7.2 Supplementary Material for Chapter 4

7.2.1 Pseudocode for Training

Algorithm 1 Decision-aware ML training

Input:
- $\{\mathbf{y_t}\}_{\mathbf{t=1}}^{\mathbf{T}}$, train data, each $\boldsymbol{y}_t \in \mathbb{R}_{\geq 0}^{S}$
- $\phi \in \mathbb{R}^P$: parameter vector for model
- M: int num MC samples for score func estimator
- J: int num MC samples for stochastic smoothing estimator
- $\sigma > 0$: float stddev of stochastic smoothing estimator
- $\epsilon \in (0,1)$: Desired minimum BPR value. Will try to enforce constraint BPR $\geq \epsilon$.
- $\lambda > 0$: Strength multiplier when constraint is violated.

Output: Trained model parameter ϕ
Procedure:

while not converged **do**
  $\nabla_\phi \mathcal{J} \leftarrow \mathbf{0}$ // $P \times 1$ vector to store grad wrt params
  for time $t \in \{1, 2, \dots T\}$ **do**
    $\{\mathbf{y}_t^m\}_{m=1}^M \sim p_\phi$ // M Monte Carlo (MC) samples
    $\mathbf{r}_t \leftarrow \frac{1}{M} \sum_m \frac{\boldsymbol{y}_t^m}{\mathbf{1} \cdot \boldsymbol{y}_t^m}$ // $S \times 1$ ranking vector via MC

    $\mathbf{b}_t \leftarrow \textsc{TopKMask}(\mathbf{r}_t)$
    $L_t^{\text{BPR}} \leftarrow -\frac{1}{\boldsymbol{y}_t \cdot \textsc{topKMask}(\boldsymbol{y}_t)} (\boldsymbol{y}_t \cdot \mathbf{b}_t)$ // Scalar loss, -BPR
    $g_t^{\text{BPR}} \leftarrow \epsilon + L_t^{\text{BPR}}$ // Scalar. Negative if BPR $\geq \epsilon$ is satisfied.
    $\mathcal{J}_t^{\text{BPR}} \leftarrow \lambda \max(g_t^{\text{BPR}}, 0)$ // Scalar ultimate loss for BPR

    $\nabla_\phi r_t \leftarrow \frac{1}{M} \sum_m [\nabla_\phi \log p_\phi(\boldsymbol{y}_t^m)] \frac{\boldsymbol{y}_t^m}{\mathbf{1} \cdot \boldsymbol{y}_t^m}$ // $P \times S$ matrix, score func. est. of $\nabla_\phi r_t$
    $\{\mathbf{z}_j\}_{j=1}^J \sim \mathcal{N}(0, I_S)$
    $\nabla_{r_t} b_t \leftarrow \frac{1}{\sigma} \frac{1}{J} \sum_{j=1}^J \textsc{outer}(\textsc{TopKMask}(\mathbf{r}_t + \sigma \mathbf{z}_j), \mathbf{z}_j)$ // $S \times S$ matrix, perturbed estimate of $\nabla_{r_t} b_t$
    $\nabla_{b_t} \mathcal{J}_t^{\text{BPR}} \leftarrow -\lambda \frac{1}{\boldsymbol{y}_t \cdot \textsc{topKMask}(\boldsymbol{y}_t)} \boldsymbol{y}_t \cdot \mathbf{1}[g_t^{\text{BPR}} > 0]$ // $S \times 1$ vector, nonzero if constraint unsatisfied.
    $\nabla_\phi \mathcal{J}_t^{\text{BPR}} \leftarrow (\nabla_\phi r_t)(\nabla_{r_t} b_t)(\nabla_{b_t} \mathcal{J}_t^{\text{BPR}})$ // $P \times 1$ vector

    $\mathcal{J}_t^{\text{NLL}} \leftarrow -\log p_\phi(\boldsymbol{y}_t)$ // Scalar ultimate loss for NLL
    $\nabla_\phi \mathcal{J}_t^{\text{NLL}} \leftarrow -\nabla_\phi \log p_\phi(\boldsymbol{y}_t)$

    $\nabla_\phi \mathcal{J} \leftarrow \nabla_\phi \mathcal{J} + \nabla_\phi \mathcal{J}_t^{\text{NLL}} + \nabla_\phi \mathcal{J}_t^{\text{BPR}}$
  end for
  $\phi \leftarrow \textsc{GradDescentUpdate}(\phi, \nabla_\phi \mathcal{J})$ // Use steepest descent or Adam or ...
end while
return ϕ

7.2.2 Ranking Demo for BPR

7.2.3 Justification for ratio estimator

In the text, we propose the ratio estimator for ranking locations: $\boldsymbol{r} = \mathbb{E}[\frac{\boldsymbol{y}}{\mathbf{1}\cdot\boldsymbol{y}}]$ and justify its use by demonstrating that it is a better bound on our decision loss BPR than the mean estimator. However, a natural question is, why not use a ranking estimator which more closely resembles BPR, such as $\boldsymbol{r} = \mathbb{E}[\frac{\boldsymbol{y}}{\text{TopKMask}(\boldsymbol{y},K)} \cdot \boldsymbol{y}]$. In practice, this estimator is equivalent to the ratio estimator in expectation, and will select the same top-K locations. Accordingly, we use the ratio estimator for its improved computational speed and simplicity.

7.2.4 Demo experiment

To gain understanding about the problem of *ranking* to optimize the fraction of best possible reach (BPR) performance metric, here we present detailed analysis of a toy problem with $S = 9$ sites. We'll assume the true data-generating process for each site is completely known throughout and that each site can be modeled independently of other sites.

$$p(\boldsymbol{y}_{1:9}) = \prod_{s=1}^{9} p(y_s) \tag{7.1}$$

For each site, we select from 3 possible site-specific model archetypes, nicknamed type A, type B, and type C.

- sites #1, #2, and #3 are each iid with type A PMF
- sites #4, #5, and #6 are each iid with type B PMF
- sites #7, #8, and #9 are each iid with type C PMF

Each type's PMF function over the non-negative integers is defined in the table below.

We have now defined the joint PMF $p(\boldsymbol{y}_{1:9})$ over the 9 sites.

	Type A	Type B	Type C
	$p(y_s) = \begin{cases} 0.0 & \text{if } y_s = 0 \\ 1.0 & \text{if } y_s = 7 \\ 0.0 & \text{otherwise} \end{cases}$	$p(y_s) = \begin{cases} 0.35 & \text{if } y_s = 0 \\ 0.65 & \text{if } y_s = 10 \\ 0.0 & \text{otherwise} \end{cases}$	$p(y_s) = \begin{cases} 0.90 & \text{if } y_s = 0 \\ 0.10 & \text{if } y_s = 80 \\ 0.0 & \text{otherwise} \end{cases}$
Mean	7.0	6.5	8.0
Median	7.0	10.0	0.0

We can compare two possible ways to compute a numerical ranking of the $S = 9$ sites:

- **Mean estimator**, which computes the per-site mean: $\boldsymbol{r} = \mathbb{E}[\boldsymbol{y}]$
- **Ratio estimator**, which computes the expectation of $\boldsymbol{y}$ normalized by its sum: $\boldsymbol{r} = \mathbb{E}[\frac{\boldsymbol{y}}{\mathbf{1} \cdot \boldsymbol{y}}]$

For each possible ranking strategy, we repeated BPR calculations over 10000 trials. To ensure accuracy of Monte Carlo estimates, we average over $M = 50000$ samples to estimate the expectation defining each $\boldsymbol{r}$.

Results are provided in the table below. We have two key findings. First, our proposed ratio estimator can select very different sites than the per-site mean, *even when both estimators have access to the true data-generating model.* Using $K = 3$, our estimator would select all type-A sites as the top 3; in contrast the per-site mean would select all type-C sites (#7-9). Second, this can produce very different BPR values. Even in this simple example, we see an absolute difference in BPR of over 0.4 between the different rankings at $K = 1$ and over 0.39 at $K = 3$, which is a huge shift for a metric that is bounded between 0.0 and 1.0.

		BPR			fraction of trials each site in top K=3 of $\boldsymbol{r}$							
		K=1	K=3	K=6	A:#1	A:#2	A:#3	B:#4	B:#5	B:#6	C:#7	C:
mean	$\boldsymbol{r} = \mathbb{E}[\boldsymbol{y}]$	0.107	0.231	0.636	0.0	0.0	0.0	0.0	0.0	0.0	1.0	
ratio	$\boldsymbol{r} = \mathbb{E}[\frac{\boldsymbol{y}}{\mathbf{1} \cdot \boldsymbol{y}}]$	0.538	0.625	0.810	1.0	1.0	1.0	0.0	0.0	0.0	0.0	

These results can be replicated via scripts provided in the code repository.

	# Spatial Sites	Temporal Scale	Outcomes $\boldsymbol{y}$	Features
MA Fatal Overdoses	1620 Census Tracts	20 years, 2001-2021	Count of opioid-related fatal overdoses	SVI, Past Deaths, Location, Time
Cook County IL Fatal Overdoses	1328 Census Tracts	8 years, 2015-2022	Count of opioid-related fatal overdoses	SVI, Past Deaths, Location, Time
Aransas TX Whooping Cranes	1338 boxes, each 500m×500m	60 years, 1952-2011 (last 10 years used)	Count of bird spotted in aerial survey	Past observations, Location, Time, Month

Table 7.1: Comparison of Real datasets, in terms of number of spatial sites S, temporal scales, outcomes $\boldsymbol{y}$, and features.

7.2.5 Experimental Details and Results from Real-World Data

7.2.5.1 Results on alternative models and metrics

7.2.5.2 Opioid-related Overdose Forecasting Results

7.2.5.3 Features

The feature vector $\boldsymbol{x}_{st}$ for site s and time t includes:

- Latitude and Longitude of the centroid of the census tract s
- The current timestep t
- Overdose gravity, an weighted average of the prior year's overdose deaths in all contiguous tracts. We construct the feature in the same way as Marks et al. [2021b]. Note that the original paper describes a weighted average over all regions within a radius, but the provided code uses only immediately contiguous locations. We follow the implementation from the code.
- Covariates from the Social Vulnerability Index (SVI) CDC ATSDR [2022]. These covariates are available as 5-year estimates for every census tract in the United States and are updated every 2 years. The SVI measures report the percentile ranking of every census tract according to 4 themes: Socioeconomic, Household Composition & Disability, Minority Status & Language, and Housing Type & Transportation. We use 5 variables: each tract's ranking in each

ANWR TX Cranes	MAE	RMSE	BPR-50
Last timestep	0.24	1.04	0.27
Avg over 10	0.25	0.78	0.38
Chance decision, $\hat{y} = 0$	0.15	0.79	0.05
EpiGNN	0.38	0.70	0.08
PG	1.03	4.85	0.35
SPO+	0.15	0.83	0.18
NLL Only	0.35	1.14	0.38
DAML ($\epsilon = 1$)	9.08	35.89	0.39
BPR Only	40495	40495	0.41

(a) ANWR TX Cranes 2009-2010.

Cook County IL Overdose	MAE	RMSE	BPR-100
Last timestep	1.07	1.66	0.76
Avg over 5	0.99	1.58	0.80
Chance decision, $\hat{y} = 0$	1.37	2.46	0.20
EpiGNN	1.33	2.03	0.32
PG	5.65	17.03	0.80
SPO+	1.25	2.38	0.74
NLL Only	1.03	1.58	0.78
DAML ($\epsilon = 1$)	1.12	1.84	0.80
BPR Only	70.31	152.72	0.82

(b) Cook County IL 2021-2022.

Table 7.2: Performance comparison between different model families for error-based and decision metrics. The top 2 rows represent simple historical baselines: using either the previous timestep alone, or an average over a larger amount (10 bi-months for the crane dataset, and 5 years for Cook County). Next, the *Chance decision* model chooses spatial locations by random chance, and uses all 0's for predicted $\hat{y}$ in MAE and RMSE calculations. Despite the simplicity, this is a competitive model on the error-based metrics due to sparsity. It outperforms all others on MAE on the Cranes dataset. Next is EpiGNN, a spatiotemporal forecasting model based on graph neural networks, trained to minimize RMSE. Although this model has the best performing RMSE on the crane dataset, it makes poor top-K decisions on both. Finally, the last three rows show different ways to train the negative binomial mixed effects regression model in 4.14. *NLL Only* trains to maximize likelihood alone (4.8), *BPR Only* is our direct-loss minimization trained to maximize BPR alone (4.9), and DAML (4.13) is our hybrid loss that allows a user to explore the pareto frontier between likelihood and BPR-K.

of the four themes as well as its composite ranking.

- We include 5 temporal lags: the number of fatal opioid-related overdoses in tract s in each of the past 5 years.

7.2.5.4 Hyperparameter ranges

Hyperparameter ranges explored include:

- Perturbation noise: $0.1, 0.01$, and 0.001.
- Adam step size: $0.1, 0.01$, and 0.001.
- Number of samples for score function trick estimator: 100
- Number of samples for perturbation estimator: 100

- BPR constraint ϵ: 5 possible values of the penalty threshold: 1.0, for a threshold that always encourages better BPR, as well as 4 values selected to be around the best BPR obtained on the training data.
- Multiplier on the penalty for DAML: 30. This value was chosen so that the BPR and likelihood components were roughly the same magnitude after 100 epochs of training.

For each location, training objective (likelihood, direct loss minimization, and DAML), as well as for each threshold for the hybrid model, we selected the model based on validation dataset performance. For the maximum likelihood model and direct BPR models we picked the model that best maximized their respective objective. For the DAML models, we found that given the larger number of hyperparameter configurations, small amount of validation data, and BPR's sensitivty, selected a model based on BPR lead to overfitting. To ameliorate this, we chose the model with the highest likelihood provided that the BPR was greater than a given threshold. Because models failed to meet the target ϵ, we chose the BPR of the maximum likelihood model on the validation dataset as our threshold. If no models met this threshold, we selected the maximum BPR model. For the We did this by evaluating the validation performance every 10 epochs, and saving a checkpoint for the model with the lowest loss on validation data.

7.2.5.5 Endangered Bird Forecasting Results

7.2.5.6 Features

The feature vector $\boldsymbol{x}_{st}$ for site s and time t includes the following variables: the past 5 count values at site s, latitude & longitude of site s's centroid, an enumerated timestep indicating time passed at t since the starting timestamp of the dataset, and a monthly indicator variable. The dataset includes three bimonthly periods per wintering season: `Oct 20-Dec 24`, `Dec 25-Feb 27`, and `Feb 28-Apr 30`. The monthly indicator received a value of 1, 2, or 3, respectively, depending on which set of months an observation spanned.

7.2.5.7 Hyperparameter ranges

The same hyperparameter and model selection was performed on Whooping Crane data as the opioid experiment in Section 7.2.5.2.

7.2.6 Experimental Details and Results from Synthetic Data

Model details. As explained in the main paper, we use a mixture of $L = 2$ Gaussian distributions where each is *truncated* to the positive reals. We give each of the S locations their own mixture weights $\pi_{s,l}$ where $\sum_{l=1}^{2} \pi_{s,l} = 1$ and $\pi_{s,l} \geq 0$. Our model for an individual location is then:

$$p(y_s) = \sum_{l=1}^{2} \pi_{s,l} \cdot \mathcal{N}_{+}(y_s|\mu_l, \sigma_l^2) \tag{7.2}$$

Our parameter vector ϕ then consists of the set of all μ_l, σ_l and $\pi_{s,l}$. We have that both μ_l and σ_l should be positive, as we are modeling positive counts and standard deviation is defined to positive. To accomplish this, we transform these variables using the softplus function when performing gradient based learning. Additionally, we constrain $\sigma_l \geq 0.2$ to avoid degeneracy. Finally, to make a valid pdf, we have that the mixture weights must sum to one: $\sum_{l=1}^{L} \pi_{s,l} = 1$. To enforce this, we transform unconstrained variables using the softmax transform. This creates an subtle issue with model identifiability, as there are many unconstrained values that will lead to similar weights, but we do not find that this impacts our ability to train models effectively to achieve good likelihood or good BPR with the appropriate objective.

Training Procedures. We seek to train 7 models: one that optimizes only for model log likelihood, one that optimizes only BPR-5, and 5 models penalized to achieve certain threshold values of BPR. Each model is given 20 random initializations of the model parameters. We use a learning step-size of 0.1. For models with BPR in the objective, we try a perturbation noise of both 0.01 and 0.05, with 500 samples. In the hybrid models, we use $\lambda = 30$, selected so that the likelihood term and the BPR penalty term are on the same order of magnitude after several epochs of training.

Tradeoffs between likelihood and BPR when L=2. If this model had $L = 7$ components, it would be well-specified and recover the true data generating process. However, with only 2 components, it is forced to group locations with distant mean values, which will come at a cost to likelihood and predictive capability as assessed by BPR.

For this example, we will consider BPR-5 as our decision making metric. A model that correctly ranks the top-5 locations will achieve perfect BPR. Our misspecified model is capable of this by learning 2 distinct mean values μ_k, one higher than the other. As long as the mixture weights for the top-5 locations π_s assign all probability to the high component, and the mixture weights for the bottom 2 locations assign all probability to the low component, the model will have perfect BPR.

However, this is not what the model with the best possible likelihood looks like. To maximize likelihood, a model will assign the 6 low locations to one component, and the one high location to another, as in Fig. 4.2.

Our hybrid objective DAML can explore the Pareto frontier between maximizing for likelihood and BPR-5. By including more locations into the high-valued component, BPR-5 will increase as log likelihood slightly decreases. Our hybrid objective formulation allows us to control this tradeoff by specifying the threshold at which the penalty term takes effect.

Results. Results are shown in Figure 4.2 in the main paper. Here we see the ability of the hybrid Decision-aware object to traverse the Pareto frontier between the best possible likelihood and BPR. The lowest threshold of 0.5 is trivially satisfied by the maximum likelihood model. The highest threshold of 1.0 is only satisfied by perfect BPR, while the 3 intermediate thresholds were chosen to explore the solution frontier that is possible by including or excluding a particular component from one of the learned mixtures. We see that by using the decision-aware training objective, we can maximize BPR while still obtaining a highly likely model.

7.2.7 Model for Negative Binomial Mixed-Effects

Here we provide further detail on the model described in Eq. (4.15).

For some elements of the parameters ϕ, we have a prior that informs point estimation in MAP fashion. The random effects are assumed to follow a multivariate normal prior

$$\begin{pmatrix} b_{0s} \\ b_{1s} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{\Sigma}\right) \tag{7.3}$$

where the covariance matrix $\mathbf{\Sigma}$ is parameterized as:

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_0^2 & \rho\sigma_0\sigma_1 \\ \rho\sigma_0\sigma_1 & \sigma_1^2 \end{pmatrix} \tag{7.4}$$

Here, σ_0 is the standard deviation of the random intercepts, σ_1 is the standard deviation of the random slopes, and ρ is the correlation between the random intercepts and slopes, where $-1 \leq \rho \leq 1$. We pack these hyperparameters into a separate vector $\eta = \{\sigma_0, \sigma_1, \rho\}$.

Transforming to unconstrained parameters. To ensure parameter constraints are satisfied throughout gradient descent optimization, we employ invertible transformations that map constrained domains to unconstrained spaces. We reparameterize the correlation coefficient $\rho \in (-1, 1)$ using $u \leftarrow \text{arctanh}(\rho)$, allowing u to be optimized over $\mathbb{R}$ while ensuring $\rho = \tanh(u)$ remains within its valid bounds. For the strictly positive parameters $\sigma_0, \sigma_1 > 0$, we optimize unconstrained parameters $\xi_0, \xi_1 \in \mathbb{R}$ and apply the softplus transformation $\sigma_i \leftarrow \ln(1 + e^{\xi_i})$, which guarantees positivity while maintaining smoothness. Similarly, for the probability parameter $q \in (0, 1)$, we optimize an unconstrained parameter $\zeta \in \mathbb{R}$ and apply the sigmoid transformation $q \leftarrow 1/(1 + e^{-\zeta})$. During gradient descent, we optimize these unconstrained parameters, and transformed to the constrained version during forward sampling or pdf evaluation of the model.

MAP estimation. Together, the parameters ϕ and hyperparameters η comprise this model. Both are point estimated to maximize the MAP objective. Thus, what is marked in the paper as NLL optimization is really best viewed as MAP or penalized NLL estimation, where the loss is

$$\mathcal{J}(\phi, \eta) = -\log p(\boldsymbol{y}_{1:T}|\boldsymbol{\phi}) - \log p(\phi|\eta) \tag{7.5}$$

Similarly, when we fit this model with DAML, the above loss is what is minimized subject to the BPR constraint.

7.3 Supplementary Material for Chapter 5

7.3.1 Synthetic Task: Further Details

In this task, we replicate a misspecified synthetic data experiment from Huang and Gupta [2024a]. Here $X \sim \text{Unif}(0, 2)$, and $Y = f^*(X) + 0.5 - \zeta$, where ζ is an exponential random variable with mean 0.5. The function $f^*(x)$ denotes the *true* mean of Y as a function of x. We set it to a piecewise linear function:

$$f^*(x) = \begin{cases} -4x + 2, & \text{for } x \in [0,\ 0.55], \\ -0.2, & \text{for } x \in [0.55,\ 2]. \end{cases} \tag{7.6}$$

Sample data are shown in in Figure 7.1. This equation is reflected about the x-axis from provided equations in Huang and Gupta, but *matches* their code.

The decision task here is to select a binary z value ($\mathcal{Z} = \{0, 1\}$) so the cost $c(z, y) = z \cdot y$ is minimized. The true data-generating process $f^*(x)$ is only above zero when $x < 0.5$. Thus, to minimize expected cost, we ideally guess $z = 0$ when $x < 0.5$ and $z = 1$ otherwise.

Model. We use a 2-parameter linear model: $\hat{y}_\theta(x) = \theta_1 x + \theta_0$. While misspecified, by design it can still yield optimal decisions. As long as the learned zero-crossing occurs at $x = 0.5$, there are many possible θ which would achieve optimal cost.

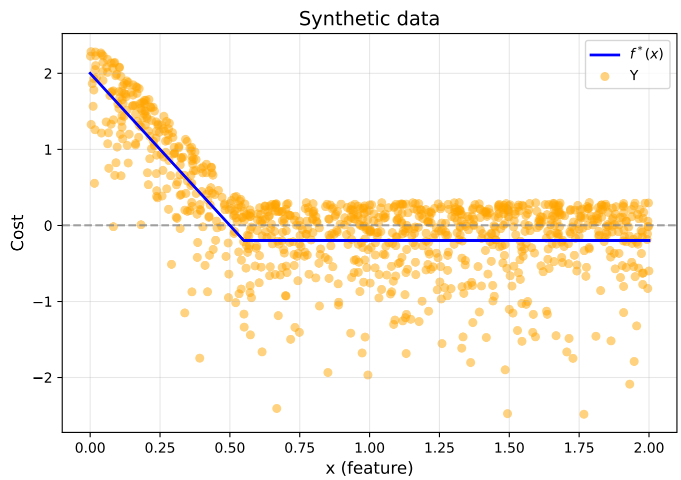

Figure 7.1: Synthetic data task: True data-generating mean function $f^*(x)$ and sampled y values using zero-mean noise from an Exponential distribution. In order to minimize cost $z \cdot y$ by taking actions $z \in \{0, 1\}$, the optimal policy is to choose z=0 when $x < 0.5$, and $z = +1$ otherwise. However, the asymmetric noise in y means that, even optimal decisions may sometimes incur a positive cost at any x value; the optimal policy's expected cost is roughly 0.39.

7.3.2 PnO 7 Task Benchmark: Further Details

7.3.3 Dataset descriptions

Knapsack using both synthetic and real energy data, **Energy Scheduling** using the same real-world dataset, **Budget allocation** based on an internet advertising dataset, **TopK selection** on synthetic data, **Bipartite Matching** on graphs of citation data, and **Portfolio optimization** on stock data.

7.3.4 Hyperparameter details

For all problems, we began with 9 training runs: a grid search of noise-level $\sigma \in [0.1, 0.5, 1]$ and learning rate $[0.01, 0.005, 0.001]$. All problems used $M = 25$ Monte Carlo samples of noise, except Energy Scheduling which used only 1 due to its long time to train.

7.3.4.1 TopK additional hyperparameter search

Following the advice of Cordonnier et al. [2021], we added a linear decay schedule to σ. We tried the following σ start and end values 0.1ß0.01, 0.1ß0.001, 0.05ß0.005, 0.05ß0.001, 0.5ß0.01, with warm-up periods of no decay lasting $0, 10$, and 30 epochs. We used the original three learning rates $[0.01, 0.005, 0.001]$, and the presence and absence of a tanh activation on the predicted costs. The best model used σ0.05ß0.001 with a warmup period of 10 epochs, no activation, and a learning rate of 0.01.

7.3.4.2 Synthetic Knapsack additional hyperparameter search

On this task, an additional $\sigma \in [0.05, 0.2, 0.3, 1]$ and learning rate ($[0.001, 0.5]$) search yielded no improvement. We attempted adding a tanh activation to bound the predicted costs, and experimented with a larger range of Monte Carlo samples $M \in [1, 5, 25, 50]$. The best model used $\sigma = 0.5$, learning rate $=$ 0.01, a tanh activation, and $M = 50$ Monte Carlo Samples.

7.3.4.3 Energy Scheduling Additional Hyperparameter Search

The model used in the Energy Scheduling task has no bounds on predicted costs, so in the second wave of hyperparameter search we tried with and without a sigmoid to constrain predicted costs in the range $[0, 1]$. We additionally attempted σ decay from the initial values to 0.01. The best model used $\sigma = 0.5$, a learning rate of 0.01, and linear decay on σ from $0.5 \to 0.01$ over 100 epochs.

7.3.4.4 Portfolio Additional Hyperparameter search

For Portfolio, we actually *removed* hyperparameters. This task has informative gradients, so there is no need for perturbation. We simply skipped the perturbation and used the best learning rate from the original grid search.

7.3.5 Top K Site Selection Task: Further Details

Figure 7.2 contains the full results from the three tasks from Heuton et al. [2025] (Predicting opioid-related fatal overdoses in Massachusetts and Cook County, IL, and forecasting Whooping Crane migration at the Aransas National Wildlife Refuge).

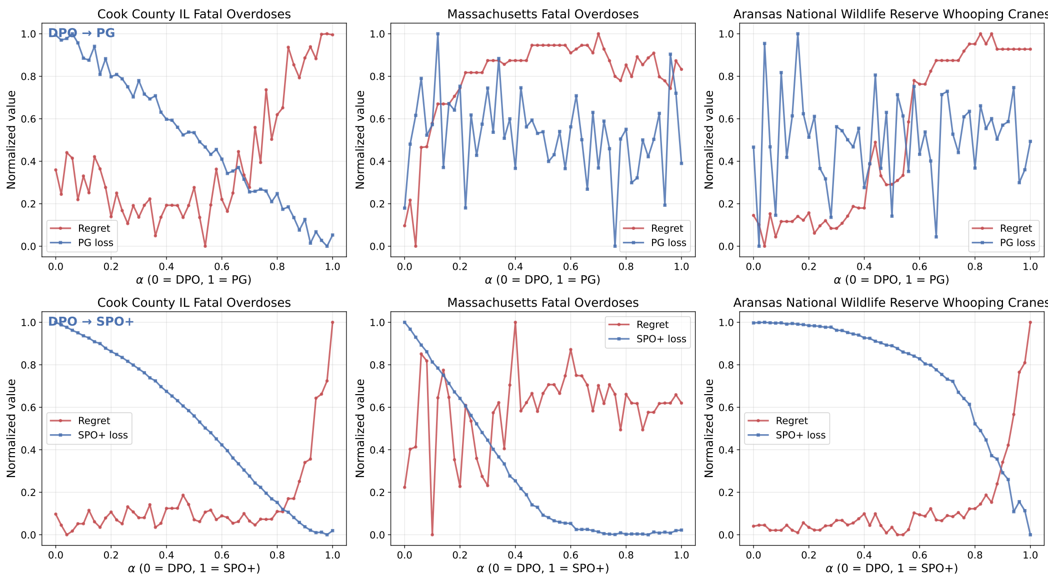

Figure 7.2: **Interpolation between DPO-learned and Surrogate-learned solutions** *Top Row:* PG loss and regret evaluated using models with parameters uniformly interpolated between the solution learned by DPO and the solution learned by PG on two public health and one wildlife datasets. *Bottom Row:* The same for SPO+. We see that the DPO loss universally learned model parameters that result in lower decision regret on the heldout dataset. Additionally, the DPO solution almost always has a higher surrogate loss value, indicating that the surrogate methods would prefer the higher regret model to the one with lower decision regret.